
Tracking Player Profile Evolution Through Robust Fuzzy Clustering: A Longitudinal Study of Transfer Market Valuation in the Primeira Liga

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Chapter 1

Introduction

Professional football is a global industry, and the transfer market is among its most visible parts. In 2019, 18,080 players moved between clubs in different countries and clubs paid USD 7.35 billion to recruit them (Franceschi, Brocard, Follert & Gouguet, 2024). Decisions of this size are still made with a good deal of judgement. Clubs overpay for players who underperform and overlook players whose value is not captured by conventional scouting, and data analytics has therefore become a tool to support more objective and replicable estimates of player value (Müller, Simons & Weinmann, 2017).

An active strand of football analytics profiles players by their performance attributes, to support scouting, squad construction and tactical planning. Clustering is particularly useful here because it groups players into profiles without relying on positional labels, which capture the role of a modern player only roughly. D’Urso, De Giovanni and Vitale (2022) propose a robust fuzzy C -medoids model for mixed-type data that handles counts, success rates and categorical positions together, assigns each player a graded degree of membership in each profile, and puts outliers into a separate noise cluster. Akhanli and Hennig (2023) develop mixed-type dissimilarities and validity indices for the same task. Together these contributions make fuzzy clustering of mixed-type performance data a rigorous way of profiling players.

Almost all of this work is cross-sectional, however. Players are clustered in a single season, so the result is a snapshot, whereas player development is dynamic: roles shift with age, club changes alter tactical contexts, and similar players follow different paths. The longitudinal literature is scarce. The closest study, by Pacifico (2025), analyses 27 representative players over twelve seasons and does not relate profile changes to market values. D’Urso et al. (2022) themselves point to the temporal dimension of the playing variables and to financial variables as natural extensions. Whether the way in which a player’s profile *changes* from one season to the next is associated with a change in his market value, once his current performance is taken into account, has not been studied. This is the question of the thesis.

1.1 Research questions and approach

The main research question is:

Do longitudinal transitions in fuzzy cluster membership, derived from a robust mixed-data clustering model applied to Primeira Liga player performance statistics across

consecutive seasons, have incremental explanatory power for changes in transfer market valuations beyond what individual performance metrics and basic player and club characteristics capture?

Two supporting questions are descriptive and methodological. The first asks which archetypes the clustering identifies, how stable they are across seasons, and how players move between them. The second asks how sensitive the answer to the main question is to the choices that have to be made when the transitions are constructed. The objective is explanatory and associational. The design is observational, no causal identification strategy is introduced, and all findings are reported as associations.

The thesis applies the model of D’Urso et al. (2022) to eight seasons (2017/18 to 2024/25) of the Portuguese Primeira Liga. Each season is clustered separately, the archetypes are aligned across seasons with the Hungarian algorithm, and the change in each player’s membership vector between consecutive seasons defines a transition. The change in the logarithm of the player’s Transfermarkt market value over the same period is then related to the transition in a regression with controls for the established determinants of value. Because the memberships are estimated, inference is based on a two-stage bootstrap that repeats the clustering and the regression (Pagan, 1984). The Primeira Liga is a deliberate choice: it is a league that supplies many players to the larger leagues, yet it has received little attention in quantitative research on player valuation, which Franceschi et al. (2024) identify as a gap.

1.2 Findings in brief

The clustering yields four interpretable archetypes of outfield players (defenders, central midfielders, wide attackers and strikers) and a small noise cluster of high-volume attackers; 75% of the players remain in their archetype from one season to the next. The estimation of the attribute weights, which is a distinguishing feature of the model, failed on these data and was replaced by weights fixed in proportion to the number of variables in each group, and the archetypes had to be aligned across seasons by their profiles instead of their medoids. In the main configuration, which was fixed before the valuation stage was estimated, membership transitions are associated with changes in market value beyond the baseline, with small gains in explanatory power, and the association is concentrated in the shift from the defender to the wide-attacker profile. The association is weak and specification-dependent, however: it is absent for some settings of the clustering, it varies strongly between random draws of the clustering, and no result survives a correction for the number of specifications examined. The thesis therefore cannot claim robust incremental explanatory power.

1.3 Relevance

Scientifically, the thesis adds a longitudinal dimension to the fuzzy clustering of mixed-type football data and links it to market valuation, thereby combining two literatures that have developed separately. It documents a failure of weight estimation in the model of D’Urso et al. (2022) and a workable alternative, and it shows how sensitive an application of fuzzy clustering

can be to its tuning. For practice, the question is whether the trajectory of a player's profile carries information about his value that scouts and sporting directors can use. The answer given here is a cautious one: the archetypes are a useful description, but their changes are, at most, weakly informative about value and not yet a reliable basis for valuation.

1.4 Structure

Chapter 2 reviews the literature on player valuation, fuzzy clustering and the clustering of football players, and states the research questions. Chapter 3 develops the theoretical argument and the hypothesis. Chapter 4 describes the data and Chapter 5 the methods. Chapter 6 reports the results for the main configuration, and Chapter 7 their sensitivity. Chapter 8 discusses the findings, implications and limitations, and Chapter 9 concludes.

Chapter 2

Literature Review

This chapter reviews the three bodies of work on which the thesis builds: the economics of player valuation (Section 2.1), fuzzy clustering methodology (Section 2.2), and the application of clustering to football players (Section 2.3). Section 2.4 identifies the gap between them and discusses the alternatives to the approach adopted here, and Section 2.5 states the research questions.

2.1 Player valuation and the transfer market

The transfer market has become an industry in its own right. In 2019, 18,080 players moved between clubs in different countries, and clubs paid a total of USD 7.35 billion to recruit them (Franceschi et al., 2024). Since the Bosman ruling, players have been treated as depreciable intangible assets when clubs invest in their recruitment, which makes the valuation of players a question of financial as well as sporting importance (Franceschi et al., 2024).

Franceschi et al. (2024) provide the most comprehensive synthesis of the empirical literature on this question. Their systematic review covers 29 peer-reviewed studies and 111 model specifications, and it classifies the explanatory variables into six categories: time, labour (for instance contract length), performance, player characteristics, club characteristics, and popularity. Several regularities emerge. Age is a recurrent regressor with a hump-shaped relationship to value: among the specifications in which the terms are significant at the 5% level, the linear term is positive in 32 of 45 and the squared term is negative in 41 of 45, so that value rises with age up to a turning point and falls afterwards. Goals and assists are positively associated with value in every specification in which they are significant, as is playing time, and playing position is a standard control. The authors also warn that the abundance of on-the-field performance variables creates potential multicollinearity, because absolute performance metrics all depend on playing time. Finally, they argue that valuation research should look beyond the five major European leagues: the Portuguese Liga is named among the leagues that had been studied in only one article, even though Benfica and Porto are, in their assessment, very important actors in player development and in the transfer market.

Most of this literature models either transfer fees or market values published by Transfermarkt, a platform on which a community of users estimates the value of players. Müller et al. (2017) use multilevel regression on 4,217 players from the top five European leagues over six

seasons to show that data-driven estimates of market value can be comparably accurate to the crowd's. Their discussion of the crowd's limitations is directly relevant to the data used here. Transfermarkt usually updates a player's value only every six to twelve months, so the estimates do not follow performance closely. In addition, crowd estimates tend to be more accurate for players who are known to a large audience, and the activity of the community is lower in some countries and leagues, which limits their usefulness for scouting players in smaller leagues. Poli, Besson and Ravenel (2024) estimate a multiple linear regression on 8,389 transfers and reach an R^2 of 84.8%, drawing on the same families of variables, among them the age of the player, the characteristics of the selling and the buying club, and the length of the remaining contract.

Two features of this literature matter for the present thesis. First, these studies relate the *level* of a player's value or fee to his current characteristics. None of them asks whether the way in which a player's multivariate performance profile *changes* is informative about changes in his value once current performance is controlled for. Second, the evidence is concentrated on the big five leagues. A league such as the Primeira Liga, which supplies many players to larger leagues, has received little attention.

2.2 Fuzzy clustering

Cluster analysis partitions observations into groups that are internally similar. Hard algorithms such as k -means assign every observation to exactly one cluster, which imposes a sharp boundary even where the data show none. Fuzzy clustering relaxes this requirement by assigning each observation a degree of membership in every cluster; memberships lie between zero and one and sum to one over the clusters. The Fuzzy C-Means (FCM) algorithm of Bezdek, Ehrlich and Full (1984) minimises a generalised least-squares objective by alternating between updating the cluster prototypes and the memberships. Its behaviour is governed by the fuzziness exponent $m > 1$: values close to one yield almost crisp partitions, whereas larger values spread the membership more evenly across clusters. In practice convergence is reached in 10 to 25 iterations, and partition-based indices such as the partition coefficient and the partition entropy are used to judge a solution (Bezdek et al., 1984).

Three extensions are relevant here. First, in *fuzzy C-medoids* the prototype of each cluster is an actual observation rather than an average. This allows dissimilarities that are not Euclidean, such as those needed for mixed-type data, and it makes the prototypes interpretable: each is a real player (Krishnapuram, Joshi, Nasraoui & Yi, 2001). For medoid-based methods, low values of m between one and about 1.5 are recommended, because with larger values the objective becomes insensitive to the choice of the medoids (D'Urso et al., 2022). Second, *noise clustering* adds an extra cluster that lies at a fixed distance from every observation, so that observations far from all genuine clusters obtain most of their membership in the noise cluster and do not distort the others (Davé, 1991). Third, a validity index is needed to compare solutions with different numbers of clusters. The index of Xie and Beni (1991) relates the compactness of the clusters to the separation of their prototypes and is used by D'Urso et al. (2022).

2.3 Clustering football players

Spatial and tactical clustering. Behravan, Zahiri, Razavi and Trasarti (2019) cluster the locations from which players are active, using a particle swarm optimisation algorithm that also determines the number of clusters. Their data consist of about 93,000 objects representing the centres of the performance of players in roughly 4,900 matches in European leagues, and the method recovers positional roles. The approach uses position data only and ignores performance statistics. Narizuka and Yamazaki (2019) work at the level of the team: they describe the formation of a team in a given moment as an adjacency matrix obtained from a Delaunay triangulation of the player positions and apply hierarchical clustering to these matrices, using player tracking data from Japanese league matches. Their method makes it possible to follow how a team’s formation changes within and across matches, but it requires tracking data, which are not available for most leagues and seasons, and it does not profile individual players.

Mixed-type clustering of performance profiles. Player performance data contain counts, rates and categorical information, so the choice of dissimilarity is central. Akhanli and Hennig (2023) analyse the 2014–15 season of eight major European leagues (England, Spain, Italy, Germany, France, Russia, the Netherlands and Turkey). Of 3,003 players who appeared at least once, goalkeepers were excluded and 1,501 outfield players with sufficient playing time were analysed. They combine group-wise dissimilarities in which each group of variables is standardised by the standard deviation of its dissimilarities and weighted proportionally to the number of variables in the group, and they select among clustering methods and numbers of clusters with a composite index that aggregates several calibrated validity criteria. A survey among 13 football experts informs the weighting of the criteria for a second, much finer clustering with 100 to 150 clusters, intended for scouting; a coarse solution with five clusters serves broad classification.

D’Urso et al. (2022) propose the model on which this thesis is built, a robust fuzzy C -medoids clustering model for mixed data with a noise cluster (FCMd-MD-NC). They group the variables into types, use an appropriate dissimilarity for each type, and estimate the weight of each type during the optimisation. Applied to the 2018/19 Serie A season, with data from WhoScored, they retain 397 of 544 players, namely those with at least 200 minutes played, because per-90-minute statistics based on fewer minutes are unreliable. The best solution has three clusters and a noise cluster that contains 49 players, with $m = 1.3$. They show that fuzzy memberships reveal intermediate players between the archetypes. They note that profiles of this kind are relevant for clubs in team selection and in determining the value of a player in the transfer window, and their concluding remarks point to extensions that include the temporal dimension of the playing variables and financial variables of the player and the team.

Longitudinal work. The study closest to a longitudinal analysis is that of Pacifico (2025). The author assembles, by manually reconciling statistics from Transfermarkt, FBref and WhoScored, a panel of 27 representative players who featured in European competitions and for their national teams, over the twelve seasons from 2012/13 to 2023/24, with 35 numerical variables; the data are described as a synthetic dataset. A fuzzy clustering of offensive indicators yields

two profiles of mostly forwards, and a gradient boosting model with interpretability tools is used to predict whether a player exceeds a goals-per-shot threshold. The paper suggests that updating the data each season would allow analysts to monitor players' cluster transitions, but it does not analyse such transitions and does not relate them to market valuations. Salary enters only as one predictor among others.

2.4 The longitudinal gap and alternative approaches

The studies above establish fuzzy clustering of mixed-type performance data as a rigorous way of profiling players, but with the exception of the small panel just described, each analyses a single season. A single season delivers a snapshot, whereas value is formed over time: clubs pay for expected future contributions, and the trajectory of a player may therefore be informative in a way that the current profile is not. At the same time, the valuation literature in Section 2.1 treats the profile through a list of individual statistics. The combination of the two, namely the movement of a player through a space of performance archetypes and its association with changes in market value, has not been studied. This is the gap addressed by the thesis, and it coincides with the extensions suggested by D'Urso et al. (2022).

Several alternative strategies could in principle address the gap, and it is worth stating why the approach adopted here is preferred. *Time-series clustering* treats each player as one multivariate time series and clusters the whole trajectories; fuzzy versions allow a series to switch between clusters (D'Urso & Maharaj, 2009). This answers a different question, namely which players follow similar paths, and it assigns each player a single cluster for the whole period. It needs reasonably long and complete series, whereas the data here are an unbalanced panel of at most eight seasons in which players enter and leave the league. *State-space or hidden Markov formulations* would model the evolution of a latent profile explicitly, but they rest on parametric assumptions about the dynamics that are difficult to justify with annual observations. *Dimensionality reduction*, for example principal component analysis, would summarise the performance statistics in a few scores whose changes could be related to value. The scores are linear combinations of the same statistics, they do not produce interpretable archetypes, and they do not allow for partial membership; the thesis nevertheless uses them as an empirical benchmark (Chapter 7). The approach adopted here clusters each season separately, aligns the archetypes across seasons, and takes differences of the membership vectors. Its price is that it ignores temporal dependence in the estimation of the profiles; its advantages are transparency, direct comparability with the cross-sectional literature, and an interpretation in terms of players and archetypes that practitioners can follow.

2.5 Research questions

The thesis pursues one main research question and two supporting questions. The main question is:

Do longitudinal transitions in fuzzy cluster membership, derived from a robust mixed-data clustering model applied to Primeira Liga player performance statistics across

consecutive seasons, have incremental explanatory power for changes in transfer market valuations beyond what individual performance metrics and basic player and club characteristics capture?

The objective is explanatory and associational. The design is observational, no causal identification strategy is introduced, and findings are reported as associations. Explanatory power is assessed in sample, with tests of the added regressors and the gain in adjusted R^2 , and out of sample, with the gain in cross-validated R^2 as a diagnostic of predictive relevance. The two supporting questions are descriptive and methodological:

- RQ1. Which player archetypes does a robust mixed-data fuzzy clustering identify in the Primeira Liga, how stable are they across seasons, and how do players move between them?
- RQ2. How sensitive is the answer to the main question to the choices that have to be made when constructing the transitions, in particular the number of clusters, the fuzziness of the memberships, the treatment of rate variables, the noise cluster, the choice of baseline, and the use of estimated rather than observed memberships in the regression?

Chapter 3

Theoretical Framework

This chapter sets out why the movement of a player through a space of performance archetypes might be associated with changes in his market value beyond what individual statistics capture, and why it might not. It also fixes the hypothesis, the interpretation of the coefficients, and the logic of the baselines against which the added explanatory power is judged.

3.1 Memberships as soft roles

Fuzzy clustering assigns each player-season (i, t) a membership vector $u_{it} = (u_{i1t}, \dots, u_{i,K+1,t})$ over K substantive archetypes and one noise cluster. The entries lie between zero and one and sum to one. The vector is the primary object of analysis, and it should be read for what it is: a set of continuous *similarity scores* that describe how closely the statistical profile of a player in a season resembles each archetype, represented by an actual player, the medoid. It is not a probability that the player “is” of a given type, nor a latent trait in a structural sense. A membership of 0.7 in one archetype and 0.3 in another says that the season’s statistics are much closer to the first but retain a clear affinity with the second. This reading suits football, in which roles are rarely exclusive: a midfielder may defend and carry the ball forward in the same season. It also explains the role of the noise cluster. A player whose profile is far from every archetype, for example a forward with an exceptional shot volume, receives most of his membership there.

3.2 Why transitions may be informative

A *membership transition* is the change in the membership vector between two consecutive seasons, $\Delta u_{it} = u_{i,t+1} - u_{it}$. It records the direction and the size of a player’s movement through the archetype space. The claim that such movements are associated with changes in market value beyond individual statistics rests on three arguments.

Forward-looking valuation. A market value is an estimate of the present value of what a player is expected to contribute in the future, not a reward for past output. The direction of a player’s movement, for instance towards a more attacking or a more central role, may inform expectations about the persistence of his output in ways that a single season’s statistics do

not. Two players with the same goals and assists may be moving in different directions, one consolidating a role and the other drifting away from one.

Role-specific valuation. Position is a standard control in valuation studies (Franceschi et al., 2024; Poli et al., 2024), because markets price roles differently. A change of role changes the group of players against which a player is evaluated. The change in value that follows a move from one archetype to another need not equal what the change in his raw statistics would imply under a linear specification.

Nonlinear aggregation. Memberships are nonlinear, low-dimensional summaries of many statistics: they depend on distances to prototypes, on all variables jointly, and on the relative position of a player among the other players of the season. Their changes can therefore reflect combinations and interactions of changes in several statistics that a linear specification with individual statistics does not represent.

3.3 Why they may not be

There are also reasons to expect little. First, memberships are deterministic functions of the same performance statistics, together with positions, that valuation models already use. Any information they contain beyond a linear specification must therefore come from the nonlinear way in which they combine those statistics, and a sufficiently flexible specification of the statistics could absorb it. Second, memberships are *estimated* quantities. The difference of two estimated memberships is noisier than either, and the use of estimated regressors in a second stage has two consequences: conventional standard errors ignore the first-stage uncertainty (Pagan, 1984), and noise in a regressor attenuates its coefficient towards zero. Third, the outcome is measured with considerable noise. Transfermarkt values are crowd estimates that are updated only every six to twelve months and are thinner for smaller leagues (Müller et al., 2017), so real changes in value are recorded with delay and in steps. Finally, values respond to many things that are not observed here, such as contract length, injuries and international appearances (Franceschi et al., 2024; Poli et al., 2024).

3.4 Hypothesis and interpretation

Let $\Delta \ln V_{it}$ denote the change in the logarithm of a player’s market value between the end of season t and the end of season $t + 1$, and let z_{it} be a vector of controls. The conceptual model is

$$\Delta \ln V_{it} = z'_{it}\beta + \Delta u'_{it}\theta + \varepsilon_{it}. \quad (3.1)$$

Because the memberships of a player sum to one in each season, the elements of Δu_{it} sum to zero, and one element has to be omitted from (3.1). The omitted element defines a *reference archetype*, which is the most common archetype in the sample. Each coefficient in θ then measures the association of the change in value with shifting membership from the reference archetype to the archetype in question, holding the controls fixed. Because a full shift of one unit is never

observed in practice, coefficients are read for a shift of 0.10: a coefficient θ_k corresponds to a proportional change in market value of approximately $100(e^{0.1\theta_k} - 1)$ per cent. The coefficients are comparable across archetypes only relative to the common reference.

The hypothesis concerns the block of coefficients and not individual archetypes, because there is no prior reason to expect a particular sign for each:

$$H_0 : \theta = 0 \quad \text{against} \quad H_1 : \theta \neq 0, \quad (3.2)$$

given the controls. A rejection indicates that membership transitions carry explanatory information about changes in value over and above the controls, and a failure to reject indicates that they do not. In both cases the statement is associational.

3.5 Baselines and the overlap with performance statistics

The memberships are constructed from twelve performance statistics, which are also the natural controls in a valuation model. The test in (3.1) is therefore only informative if the controls are chosen with care. Two baselines are used. The *core* baseline follows the determinants that are most consistently reported in the literature (Franceschi et al., 2024): age and its square, playing time and its change, playing position, the points per match of the team while the player is on the pitch and their change, an indicator of a change of club within the league, season effects, and goals and assists per 90 minutes in their level and in their change. The *full* baseline adds the level and the change of all twelve statistics from which the memberships are built. Against the full baseline, the transitions can contribute only through the nonlinear way in which they combine these statistics, which is exactly the claim in Section 3.2. A linear summary of the statistics cannot add anything to the full baseline by construction, since it is a linear combination of variables that the baseline already contains; principal component scores are therefore compared with the transitions against the core baseline only (Chapter 7).

Chapter 4

Data

The analysis combines two sources linked at the player level: performance statistics from FBref and market valuations from Transfermarkt, for the eight Primeira Liga seasons from 2017/18 to 2024/25. This chapter describes the sources, the construction of the sample, the variables, the link between the sources, and the measurement issues that matter for the interpretation of the results.

4.1 Sources and collection

Performance data. Season-level player statistics were collected from FBref with the Python library `soccerdata`. The Primeira Liga is not among the leagues that the library supports by default, so it was registered through the library’s custom league configuration. Four statistical tables were collected for each season: standard statistics, shooting, playing time, and miscellaneous statistics. They were merged on season, club and player, which gives 4,463 club-level player-season rows for 2,008 distinct player names. A player who changed club during a season appears once for each club. FBref’s advanced Opta-based statistics, such as expected goals, expected assists and progressive passes, are not available in the tables collected for the Primeira Liga over this period, and the tables do not contain passing accuracy, dribbling or duel statistics. The set of performance variables is therefore restricted to what is available: goals, assists, shots, shots on target, crosses, interceptions, tackles won, fouls committed and drawn, offsides, and playing time. The consequence is that archetypes are defined by attacking output, defensive actions and positions, and not by passing or ball-progression behaviour. This limitation is returned to in Chapter 8.

Market values. Transfermarkt provides, for every player, a history of market value estimates. The histories were obtained from the same data service that feeds the value chart on each player’s page. To identify the players of each season, the squad lists of the 18 clubs of each of the eight seasons were collected from Transfermarkt, which gives a player identifier, a name and a birth year for every club-season. All requests were spread over time and all pages were stored locally, so that the collection can be reproduced without repeated queries.

4.2 Sample construction

The unit of observation is a player-season. The sample is built in six steps, which are shown with their counts in Table 4.1.

Table 4.1: From the collected data to the analysis sample

Step
Club-level rows collected from FBref
After removing rows of players who share a name with another player of the same club and season (22 rows, 4 groups)
One row per player and season (rows of players who changed club merged)
After removing one merged player-season with more than 3,100 minutes
After removing goalkeepers (354 player-seasons)
After removing player-seasons with fewer than 200 minutes (808)
After removing player-seasons with missing shooting or miscellaneous statistics (9)

First, the four FBref tables are joined on season, club and name, and this join cannot separate two players with the same name at the same club in the same season: each of them is paired with each row of the other tables, so that counts and minutes are duplicated and mixed (one such pair of players appeared with 22,152 minutes, whereas a 34-match season has at most 3,060). The 22 rows concerned, of four such groups, cannot be repaired from the merged data and are removed. Second, the rows of players who played for more than one club in a season (77 player-seasons) are merged: counts are summed, and per-90-minute figures and rates are recomputed from the totals, not averaged. A merged player-season with more than 3,100 minutes, which can only arise when two different players with the same name and birth year at different clubs are merged, is removed. Third, goalkeepers are removed, because their performance is of a different nature and would dominate any clustering of outfield players, as in Akhanli and Hennig (2023). Fourth, player-seasons with fewer than 200 minutes are removed, following D’Urso et al. (2022), because per-90-minute statistics based on little playing time are unreliable. Fifth, nine player-seasons, all in 2017/18, are removed because FBref has no shooting and miscellaneous statistics for them. The final sample has 3,192 player-seasons of 1,499 players, between 384 and 415 per season (Table 4.2). Players appear in 2.1 seasons on average, and more than 800 of them in at least two.

Table 4.2: Sample by season

Season	Player-seasons	Players	Share with value	Season pairs used
2017/18	384	384	0.96	206
2018/19	384	384	0.94	185
2019/20	405	405	0.91	202
2020/21	394	394	0.95	199
2021/22	395	395	0.96	198
2022/23	415	415	0.95	199
2023/24	406	406	0.94	175
2024/25	409	409	0.94	0
Total	3192	1499	0.94	1364

Note. Share with value is the share of player-seasons with a Transfermarkt value at the end of the season. Season pairs used are the pairs of consecutive seasons, starting in the season shown, for which the player qualifies in both seasons and has a value at both ends; they form the estimation sample of the valuation stage.

4.3 Variables

The clustering uses three types of attributes, which follow the grouping of variables in D’Urso et al. (2022). Table 4.3 lists them.

Table 4.3: Attributes used in the clustering

Type	Variables	Dissimilarity
Performance (10)	Non-penalty goals, assists, shots, shots on target, crosses, interceptions, tackles won, fouls committed, fouls drawn, off-sides; all per 90 minutes	Manhattan distance on scaled variables
Success rates (2)	Shots on target per shot, non-penalty goals per shot (both shrunk, see text)	Manhattan distance on scaled variables
Position (3)	Indicators for defender, midfielder and forward; a player can hold several	Jaccard dissimilarity

Per-90 statistics and scaling. Counts are divided by the number of 90-minute periods played. All numerical variables are scaled by their average absolute deviation from the median, the standardisation used by Akhanli and Hennig (2023), which is less sensitive to the heavy right tail of count statistics than the standard deviation. The scaling constants are computed once on the pooled sample of all seasons and not season by season. This keeps the units the same in every season, which is what makes medoids of different seasons comparable when the archetypes are aligned (Section 5.2).

Success rates. A rate based on few attempts is very noisy: a player with one shot and one goal has a conversion rate of 100%, and a player without shots has none. To prevent such cases from dominating, rates are shrunk towards the pooled rate,

$$\tilde{r}_{it} = \frac{s_{it} + \kappa \bar{r}}{n_{it} + \kappa}, \quad (4.1)$$

where s_{it} is the number of successes, n_{it} the number of shots, $\bar{r}$ the pooled rate over all player-seasons, and $\kappa = 10$ shots. A player without shots receives the pooled rate, which expresses that nothing is known, instead of a rate of zero. This is an empirical-Bayes type of shrinkage (Efron & Morris, 1975). The rates computed from the raw counts are used in a robustness analysis (Chapter 7). Goals per shot on target is not used because it equals goals per shot divided by the shot-on-target rate and would count the same information twice.

Positions. FBref assigns each player up to three of the labels defender, midfielder and forward. For players with several clubs in a season the union of the labels is used. Five combinations account for almost all player-seasons (defender, midfielder, forward-midfielder, forward, defender-midfielder).

Table 4.4 reports descriptive statistics. A typical player-season has about 1.3 shots, 0.11 non-penalty goals and 0.08 assists per 90 minutes, and the distributions are skewed to the right, as expected for count data.

Table 4.4: Descriptive statistics

Variable	Mean	SD	Min	Median	Max	N
Non-penalty goals per 90	0.11	0.16	0.00	0.05	1.90	3192
Assists per 90	0.08	0.12	0.00	0.04	1.06	3192
Shots per 90	1.26	0.98	0.00	0.97	7.68	3192
Shots on target per 90	0.42	0.43	0.00	0.28	3.53	3192
Crosses per 90	1.82	1.95	0.00	1.13	12.87	3192
Interceptions per 90	1.04	0.67	0.00	0.96	4.56	3192
Tackles won per 90	1.05	0.58	0.00	0.98	4.20	3192
Fouls committed per 90	1.61	0.72	0.00	1.50	8.57	3192
Fouls drawn per 90	1.48	0.91	0.00	1.34	5.87	3192
Offsides per 90	0.21	0.31	0.00	0.08	2.44	3192
Shots on target rate (shrunk)	0.33	0.07	0.12	0.32	0.92	3192
Goals per shot (shrunk)	0.09	0.04	0.01	0.08	0.30	3192
Age (years)	25.17	3.94	17.00	25.00	41.00	3192
Minutes played	1350.58	777.53	200.00	1270.00	3060.00	3192
Team points per match (on pitch)	1.36	0.59	0.00	1.21	3.00	3190
Change in log market value (t to t+1)	0.07	0.63	-1.61	0.00	3.08	1364
Size of the membership shift	0.36	0.33	0.00	0.26	1.38	1433

Note. Statistics of the clustering variables are in their original, unscaled form and refer to the 3,192 player-seasons. The change in log market value and the size of the membership shift refer to the season pairs of the estimation sample and are described in Chapter 6.

4.4 Linking the two sources

FBref and Transfermarkt use different identifiers, so players have to be linked by name and birth year. The link is made *within each season*, which keeps the pool of candidates small: for a player-season in FBref, the candidates are the players on the Transfermarkt squad lists of that season whose birth year differs by at most one year. A candidate is accepted if the similarity of the normalised names exceeds a threshold that depends on how informative the birth year is: 70 on a scale up to 100 when the birth years agree, 85 when they differ by one year, and 92 when a birth year is missing. A player is assigned the Transfermarkt identifier that is selected most often across his seasons. Of the 1,507 outfield players, 1,446 (96.0%) could be linked, which corresponds to 96.2% of the player-seasons; no player-season of the unlinked players has a value.

The linking was checked in three ways, all of which are automatic except the last. First, 32 links with a name score below 92 were listed with the names, birth years and clubs on both sides and inspected manually. Most are nicknames or short forms of the same name at the same club (for example *Oriol* and *Uri* Rosell, or *Álex* and *Alejandro* Grimaldo). One link was rejected because it joined two different players with similar surnames; two others had already been removed in the next step. Second, fifteen Transfermarkt profiles had been linked to two different FBref spellings. In twelve of them both spellings appear in the sample. For eleven of these the two spellings occur in the same season or are clearly different names, which means that they are different players (a nickname such as *Ricardo* and the full name *Ricardo Esgaio*, for example), and the weaker link was discarded so that no player receives another player's values. In one case the spellings never occur in the same season, the names are similar and the birth years are within one year of each other, so the two spellings were treated as one player. After these steps, 1,500 name and birth-year combinations are 1,499 distinct players. Players without

a link remain in the clustering sample but cannot enter the valuation stage.

4.5 Market values and the outcome

For each player and season, the value recorded closest to 1 June of the year in which the season ends is used, provided it lies within 120 days of that date; this approximates the end-of-season valuation. A value is available for 94.5% of the player-seasons (between 91% and 96% depending on the season). The outcome of the valuation stage is the change in the logarithm of the value between the end of season t and the end of season $t + 1$,

$$\Delta \ln V_{it} = \ln V_{i,t+1} - \ln V_{it}. \quad (4.2)$$

It is observed only if the player qualifies (at least 200 minutes) in both seasons, so the transitions of players who leave the league, or who play too little, are not observed. There are 1,433 such pairs, and 1,364 of them have a value at both ends; three more are dropped because the team-level control is missing, which leaves 1,361 pairs of 697 players for the estimation. The mean change is 0.07 (about 7%) with a standard deviation of 0.63; in about 14% of the pairs the value is unchanged, and in about 22% the player changed club within the league.

4.6 Measurement issues

Three features of the data affect the interpretation. First, market values are crowd estimates, updated at most every six to twelve months, and the crowd is thinner for players of smaller leagues (Müller et al., 2017). The outcome is therefore noisy and moves in steps, as the share of unchanged values illustrates. Noise in the dependent variable widens standard errors but does not by itself bias the coefficients. Second, the sample is selected: the transitions of players who leave the league are not observed, and the findings refer to players who stay. Third, memberships are estimated, and transitions are differences of estimated quantities; Section 5.5 deals with the consequences for inference.

Chapter 5

Methodology

The empirical analysis has two stages. In the *clustering stage*, each season is clustered separately with a robust fuzzy C -medoids model for mixed-type data, the archetypes are aligned across seasons, and membership transitions are computed. In the *valuation stage*, the change in log market value is related to the transitions, controlling for the determinants of value. This chapter describes both stages, the inference that accounts for the use of estimated memberships, and the design of the robustness and sensitivity analyses.

A main configuration was fixed before the valuation stage was estimated: shrunk success rates, $K = 4$ archetypes, fuzziness $m = 1.3$ and noise multiplier $\rho = 1.0$. The choice of K rests on clustering diagnostics only (Section 6.1), $m = 1.3$ is the value of D’Urso et al. (2022), and ρ was calibrated to the noise share (Section 5.1.3). All other configurations are reported as sensitivity analyses and are never used to replace the main one. The proposal envisaged selecting C and m with the Xie–Beni index. For m this was not done, because the index is not a reliable guide to the fuzziness exponent: across the specifications examined, its median falls from 1.06 at $m = 1.3$ to 0.99, 0.91 and 0.70 at $m = 1.4$, 1.5 and 2.0, in line with the near-uniform memberships that large exponents produce. The published value was therefore adopted. Within the main configuration the index happens to be lowest at $m = 1.4$ among the values from 1.2 to 1.5, and the dependence of the results on m turns out to matter; it is reported in full in Chapter 7.

5.1 Stage 1: robust fuzzy C -medoids for mixed data

5.1.1 Dissimilarities

A player-season i has three groups of attributes: performance statistics ($s = 1$), success rates ($s = 2$) and positions ($s = 3$), with $n_1 = 10$, $n_2 = 2$ and $n_3 = 3$ variables (Table 4.3). For each group, a dissimilarity $d_s(i, j)$ is computed: the Manhattan distance for the two groups of scaled numerical variables, and the Jaccard dissimilarity for positions, which treats the labels as sets and counts two players with identical sets as identical. To make the groups comparable, each is divided by the standard deviation σ_s of its pairwise dissimilarities over the pooled sample, $sd_s(i, j) = d_s(i, j)/\sigma_s$, which is the standardisation of Akhanli and Hennig (2023). The constants σ_s are computed once and kept fixed for all seasons. The overall squared dissimilarity combines

the groups through weights w_s ,

$$d^2(i, j) = \sum_{s=1}^3 w_s^2 \text{sd}_s(i, j)^2. \quad (5.1)$$

5.1.2 Weights

In D’Urso et al. (2022) the weights are estimated within the optimisation, in inverse proportion to the dispersion of each group around the medoids. On the data of this thesis this failed in every variant that was tried. With the scaling by the maximum used in the original paper, the two success rates took 96 to 97% of the weight with raw rates and 60 to 70% with shrunk rates (with 21 to 34% of the players in the noise cluster) for three and four clusters; with the scaling by the mean, positions took 76 to 91% of the weight in three of the four runs with three or four clusters; and with five clusters the weights collapsed onto the positions in all four variants, completely in three of them and to 90% in the fourth, with a validity index of zero in three (2023/24 season; Appendix B). The explanation is structural: the estimation rewards the group that is most compact around the medoids, and a group with few variables, or with few distinct values, is always more compact than ten continuous variables. The thesis therefore follows Akhanli and Hennig (2023), who fix the weights in advance in proportion to the number of variables of each group. Because (5.1) squares the weights, they are set proportional to $\sqrt{n_s}$ and normalised to sum to one, $w_s = \sqrt{n_s} / \sum_r \sqrt{n_r}$, so that the contribution of a group to d^2 is proportional to its number of variables. This gives $w = (0.501, 0.224, 0.275)$. The unsuccessful estimation of the weights is reported in Chapter 7 as a result in itself.

5.1.3 Memberships, medoids and the noise cluster

Let K be the number of substantive clusters, with medoids $g_1, \dots, g_K$ that are player-seasons of the same season, and let $K + 1$ denote the noise cluster. With $d_{ic}^2 = d^2(i, g_c)$ and fuzziness exponent $m > 1$, the clustering minimises

$$J = \sum_{i=1}^n \sum_{c=1}^K u_{ic}^m d_{ic}^2 + \sum_{i=1}^n u_{i,K+1}^m \delta^2, \quad \sum_{c=1}^{K+1} u_{ic} = 1, \quad u_{ic} \geq 0, \quad (5.2)$$

where δ^2 is the squared distance of every observation to the noise cluster. For given medoids, the memberships are

$$u_{ic} = \frac{(d_{ic}^{-2})^{1/(m-1)}}{\sum_{l=1}^K (d_{il}^{-2})^{1/(m-1)} + (\delta^{-2})^{1/(m-1)}}, \quad u_{i,K+1} = \frac{(\delta^{-2})^{1/(m-1)}}{\sum_{l=1}^K (d_{il}^{-2})^{1/(m-1)} + (\delta^{-2})^{1/(m-1)}}, \quad (5.3)$$

and the medoid of cluster c is the player-season h that minimises $\sum_i u_{ic}^m d^2(h, i)$. The algorithm alternates between the two updates until the medoids no longer change (at most 200 iterations). An observation whose distance to every medoid is larger than δ obtains most of its membership in the noise cluster. The noise distance is $\delta^2 = \rho \cdot \frac{1}{nK} \sum_i \sum_c d_{ic}^2$, the average squared distance of all observations to all medoids multiplied by a constant ρ , which is the formulation of Davé (1991). The formulation printed in D’Urso et al. (2022) weights the average by u_{ic}^m . In test runs,

that version led to degenerate solutions (in one setting 99% of the players were assigned to the noise cluster), so the unweighted average is used. With it, $\rho = 1.0$ gives a noise share of about 5% at $K = 4$ (the paper reports 12%), and ρ is varied in the sensitivity analysis.

Each season is clustered with ten random starts, with initial medoids drawn in the manner of *k-medoids++* (each new medoid is drawn with a probability that increases in its distance to the medoids already chosen), and the solution with the smallest value of J is kept. The fuzziness exponent is $m = 1.3$ in the main configuration. This value was selected by D’Urso et al. (2022) among 1.3, 1.5 and 2.0 and lies in the range recommended for medoid-based methods (Section 2.2); the sensitivity of the results to it is examined in Chapter 7.

5.1.4 Number of clusters

The number of clusters has to be the same in all seasons for the archetypes to be aligned, so it is chosen once for all seasons and not season by season as in the original paper. Candidate values of K from 3 to 6 are compared on the following diagnostics, each averaged over the seasons: an adaptation of the Xie–Beni index (Xie & Beni, 1991), $XB = \sum_i \sum_{c \leq K} u_{ic}^m d_{ic}^2 / (n \min_{c \neq l} d^2(g_c, g_l))$, with squared distances and the substantive clusters only; the share of players assigned to the noise cluster; the share of players whose largest substantive membership is at least 0.7 (crispness); the stability of the solution across random starts, measured by the adjusted Rand index between the best solution and the other starts; the size of the smallest archetype; and three diagnostics of the alignment across seasons that are defined below. The Xie–Beni index alone is not decisive, because it increases almost mechanically with K .

5.2 Alignment across seasons

Because the seasons are clustered independently, the numbering of the clusters is arbitrary, and the clusters of season $t+1$ have to be matched to those of season t . The matching is an assignment problem, solved with the Hungarian algorithm (Kuhn, 1955). What has to be specified is the cost of matching archetype k of season t with archetype l of season $t+1$. The proposal used the distance between the medoids. This was tried first, and it turned out to be unreliable: a medoid is a single player, and when two archetypes have similar medoids the matching can swap their labels between seasons, which turns a whole season pair of memberships into spurious transitions. A check of all 120 clustering runs of the specification grid found consistent labels in every season pair in only 30% of them (Section 7.6). The matching therefore uses the *profile* of each archetype instead: the membership-weighted mean, over all players of the season, of the twelve clustering variables and the three position indicators, each standardised over the pooled sample,

$$\bar{x}_k^{(t)} = \frac{\sum_i u_{ik}^{(t)} x_i^{(t)}}{\sum_i u_{ik}^{(t)}}, \quad (5.4)$$

and the cost is the Manhattan distance between profiles, $C_{kl} = \|\bar{x}_k^{(t)} - \bar{x}_l^{(t+1)}\|_1$. A profile averages the information of every player in the archetype and is therefore far more stable than a single medoid. The one-to-one assignment that minimises the total cost is found for each pair of adjacent seasons, and the labels are propagated forward from the first season. The noise cluster

is not aligned. The medoids are not used in the alignment, so the consistency of their positions across seasons is an independent check of it.

Three properties of the alignment are recorded. The *alignment distance* is the mean cost of the matched pairs, which is small if the same archetype exists in both seasons. A match is called *ambiguous* if the second-best alternative for the same row or column has a cost within 10% of the chosen match. The *persistence* is the share of players observed in two consecutive seasons whose largest substantive membership remains in the same archetype. The independent clustering of the seasons ignores temporal dependence, which is a limitation relative to dynamic approaches (Section 2.4); the price is a model that is simple to interpret and to compare with the cross-sectional literature.

5.3 Membership transitions

For a player observed in two consecutive seasons, the transition is $\Delta u_{it} = u_{i,t+1} - u_{it}$, with $K + 1$ elements that sum to zero. Besides the vector, two scalar summaries are used: the size of the shift, $\|\Delta u_{it}\|_2$ over the substantive archetypes, and the change in the entropy of the membership vector, $\Delta H_{it} = H(u_{i,t+1}) - H(u_{it})$ with $H(u) = -\sum_c u_c \ln u_c$. The first captures the size of the movement without its direction, the second whether the profile becomes more or less specialised.

5.4 Stage 2: valuation model

5.4.1 Specification

The estimation sample consists of the season pairs described in Section 4.5. The outcome is $\Delta \ln V_{it}$. The baseline specification is

$$\Delta \ln V_{it} = \alpha + z'_{it}\beta + \varepsilon_{it}, \quad (5.5)$$

and the extended specification adds the transitions,

$$\Delta \ln V_{it} = \alpha + z'_{it}\beta + \Delta u'_{it}\theta + \varepsilon_{it}, \quad (5.6)$$

in which the element of Δu_{it} of the reference archetype (the one with the largest average membership) is omitted, as explained in Section 3.4. The *core* controls z_{it} are age and its square, the logarithm of minutes played and its change, indicators for defenders and forwards, the points per match of the team while the player is on the pitch and their change, an indicator for a change of club within the league, season-pair effects, and non-penalty goals and assists per 90 minutes in their level and change. The *full* controls add the level and the change of the remaining ten statistics from which the memberships are built. Statistics are entered in the scaled form of Section 4.3.

5.4.2 Tests and measures of explanatory power

The central test is whether $\theta = 0$, and it is a Wald test of the block of coefficients with standard errors that are cluster-robust by player, because a player can contribute several pairs (Cameron

& Miller, 2015). Explanatory power is measured by the gain in the adjusted R^2 when the transitions are added, and by the gain in out-of-sample R^2 in repeated five-fold cross-validation in which all pairs of a player are in the same fold; the cross-validation is repeated 30 times with different splits. The out-of-sample gain is a diagnostic of predictive relevance and is not used as the test.

5.4.3 Fixed and random effects

Player-specific unobserved characteristics such as talent, reputation or injury history may be correlated with both memberships and values. The model is therefore also estimated with player fixed effects and with random effects, and the Hausman test (Hausman, 1978) decides between them. Since age increases by one year per season for every player, it is not identified together with player and season effects, and age and its square are left out of this comparison. Only players with at least two season pairs contribute to the fixed-effects estimates.

5.5 Estimated regressors and the two-stage bootstrap

The memberships are estimated, and conventional standard errors in the second stage ignore this uncertainty (Pagan, 1984). To account for it, both stages are repeated on bootstrap samples (Efron, 1979). In each replication, players are drawn with replacement, with all seasons of a player drawn together; each season of the resampled data is clustered again with the same model, K , m , ρ , weights and scaling constants and with ten random starts; the clusters are matched to the *original* solution season by season with the Hungarian algorithm, using the distance between the profiles of the archetypes, so that each archetype keeps its meaning across replications; and the transitions and the extended model are re-estimated. The standard error of each coefficient is the standard deviation of its 500 replicated estimates, and the block of coefficients is tested with a Wald statistic that uses the bootstrap covariance matrix. The bootstrap corrects the standard errors but not the attenuation of coefficients caused by noise in the memberships. In-sample R^2 gains computed on bootstrap samples are optimistic because duplicated observations are fitted closely, and they are not used.

5.6 Robustness and sensitivity design

The design of the robustness analyses follows the points where a choice has to be made. *Sample and outcome variations* repeat the test for the main configuration with a winsorised outcome (1st and 99th percentiles), only players who did not change club, only non-zero changes of value, players with at least 500 minutes in both seasons, without the COVID-affected pairs (starting in 2019/20 and 2020/21), players aged up to 30, and without players whose noise membership exceeds one half. *Alternative transition measures* replace the membership changes by the size of the shift and by the change in entropy. A *specification grid* varies the rate variables (shrunk or raw), the number of clusters (3 to 6), the noise multiplier ($\rho \in \{0.75, 1.0, 1.5\}$) and the fuzziness ($m \in \{1.2, 1.3, 1.4, 1.5, 2.0\}$), which gives 120 specifications, each with a re-run of the clustering, the alignment and the tests; the bootstrap is applied to a selection of them. Finally, a *principal*

component benchmark replaces the membership changes by the changes of the first principal components of the same twelve statistics, with the same number of dimensions. In addition, because the clustering is stochastic, the whole first stage is repeated with 30 different random seeds for each fuzziness value, and the test is repeated each time, to measure the dependence of the result on the random draw. Because many specifications are examined, no single p -value is read in isolation; the pattern across specifications is what is reported, and the main configuration remains the pre-specified result.

5.7 Implementation

All analyses are implemented in Python. The clustering model is implemented from scratch following D’Urso et al. (2022), because no library provides it; the assignment problem is solved with `scipy`, the regressions with `statsmodels` and `linearmodels`, the cross-validation with `numpy`, and the bootstrap is parallelised with `joblib`. All random numbers are generated from fixed seeds. The programs are described in Appendix A.

Chapter 6

Results

This chapter reports the results for the main configuration: shrunk success rates, $K = 4$ archetypes, fuzziness $m = 1.3$ and noise multiplier $\rho = 1.0$ (Section 5.1), estimated in one run of the clustering with a fixed random seed. It first shows how the number of archetypes was chosen (Section 6.1), describes the archetypes and the transitions between them (Sections 6.2 and 6.3), and then reports the valuation stage (Section 6.4). The clustering is stochastic, and a single run is one draw from a distribution of possible results; Chapter 7 examines how far the results depend on the choices made in the clustering stage and on the random draw, and the main configuration is not changed in the light of that analysis.

6.1 Choosing the number of archetypes

Table 6.1 compares $K = 3, \dots, 6$ for both versions of the rate variables, with every diagnostic averaged over the eight seasons. The choice rests on these diagnostics and on the profiles of the solutions, and was made before the valuation stage was estimated.

Table 6.1: Diagnostics for the number of archetypes ($m = 1.3$, $\rho = 1.0$; averages over seasons)

Rates	K	Xie–Beni	Noise	Crisp	Stability	Smallest	Align. dist.	Ambig.	Persist.
Shrunk	3	0.82	10.5%	0.63	0.49	22.3%	3.29	4	0.72
Shrunk	4	0.72	5.3%	0.63	0.51	13.8%	2.39	2	0.75
Shrunk	5	1.35	4.4%	0.55	0.51	11.1%	3.34	12	0.53
Shrunk	6	1.24	3.4%	0.55	0.51	8.5%	3.63	16	0.53
Raw	3	0.62	9.2%	0.66	0.50	23.7%	2.44	0	0.79
Raw	4	0.82	5.4%	0.64	0.52	13.3%	2.21	2	0.73
Raw	5	1.33	5.5%	0.55	0.50	12.6%	2.74	9	0.60
Raw	6	1.64	3.3%	0.54	0.48	7.2%	3.13	12	0.58

Note. Noise is the share of players whose largest membership is in the noise cluster; Crisp is the share with a largest substantive membership of at least 0.7; Stability is the adjusted Rand index between the best and the other random starts; Smallest is the share of the smallest archetype; Align. dist. is the mean Manhattan distance between the standardised profiles of matched archetypes in adjacent seasons; Ambig. is the number of ambiguous matches over the seven season pairs; Persist. is the share of players who keep their largest archetype in the next season. Each row is based on 1,433 transitions.

Three of the diagnostics deteriorate steeply beyond $K = 4$. Persistence falls from 0.75 at $K = 4$ to 0.53 at $K = 5$ and at $K = 6$ (shrunk rates), the number of ambiguous alignments rises from 2 to 12 and 16, and the Xie–Beni index, which is lowest at $K = 4$ among the four

values (0.72), rises to 1.35 and 1.24. The medoids show why: in the runs with five clusters, one archetype is represented by players of different types in different seasons, so that its meaning drifts. With $K = 3$ the noise cluster takes 10.5% of the players, twice as many as at $K = 4$ (5.3%), which is below the 12% reported by D’Urso et al. (2022), and persistence is no better (0.72). At $K = 4$ the smallest archetype contains 14% of the players, and the medoids remain consistent across seasons (Table 6.3). The two versions of the rate variables behave similarly; the shrunk rates give slightly higher persistence and the lowest Xie–Beni index at $K = 4$ (0.72 against 0.82), and they are preferred on the grounds given in Section 4.3. The start stability, an adjusted Rand index of about 0.5 at all K , is moderate. It reflects that player profiles form a continuum with no sharp natural boundaries, which is the case for fuzzy memberships, but it also means that individual memberships are measured with noise and that different random starts give somewhat different partitions, a point that returns in the valuation stage and in Section 7.7.

6.2 The archetypes

Table 6.2 describes the four archetypes and the noise cluster through the membership-weighted means of the performance statistics and of the position indicators, and Table 6.3 lists the medoid players. The names of the archetypes are labels chosen here to summarise what the profiles show; the model does not produce them.

Table 6.2: Archetype profiles (membership-weighted means)

Profile	Share	Goals	Assists	Shots	Crosses	Intercep.	Tackles	SoT rate	Goals/shot	DF	MF	FW
Defenders	33.0%	0.03	0.04	0.52	1.22	1.46	1.07	0.24	0.061	0.91	0.19	0.03
Strikers	14.8%	0.28	0.10	2.32	1.28	0.37	0.63	0.39	0.123	0.02	0.41	0.89
Wide attackers	19.2%	0.13	0.13	1.71	3.02	0.74	0.99	0.33	0.077	0.09	0.87	0.58
Central midfielders	27.6%	0.06	0.07	0.98	1.82	1.18	1.33	0.26	0.057	0.21	0.91	0.14
Atypical attackers (noise)	5.4%	0.29	0.19	2.36	2.41	0.69	0.97	0.38	0.125	0.13	0.55	0.61
All players		0.11	0.08	1.26	1.82	1.04	1.05	0.30	0.077	0.38	0.57	0.33

Note. Statistics are per 90 minutes (non-penalty goals, assists, shots, crosses, interceptions, tackles won) in their original unscaled form; SoT rate is shots on target per shot; DF, MF and FW are the weighted shares of players with the corresponding position label. Share is the proportion of player-seasons whose largest membership is in the profile. Rates are the raw values, not the shrunk values used in the clustering.

Defenders (33% of the player-seasons) combine the lowest shot volume (0.52 per 90) with the highest number of interceptions (1.46), and 91% of their weight is on the defender label. *Central midfielders* (28%) have the highest number of tackles won (1.33) with intermediate interceptions (1.18) and a modest attacking output; their weight is 91% on the midfielder label. *Wide attackers* (19%) are characterised by crosses (3.02 per 90, against 1.82 for all players), a high shot volume (1.71) and assists (0.13), and they combine the midfielder (87%) and forward (58%) labels. *Strikers* (15%) have the highest goal rate (0.28 per 90), the highest shot volume (2.32), and the best finishing (0.39 of shots on target and 0.12 goals per shot), with a weight of 89% on the forward label. The *noise* cluster (5%) collects high-volume attackers: the profile has 2.36 shots, 2.41 crosses, 0.29 goals and 0.19 assists per 90 minutes, all among the highest, together with a mixed position label. These are players whose output does not resemble any single archetype, such as forwards with very high shot volumes.

Table 6.3: Medoid players by season

Season	Defenders	Strikers	Wide attackers	Central midfielders
2017/18	Flávio Ramos	Alexandre Guedes	Fábio Espinho	Vítor Gomes
2018/19	Ivanildo Fernandes	Bryan Róchez	Jorge Correa	Filipe Augusto
2019/20	Tomás Ribeiro	Iury de Castilho	Silvestre Varela	Julian Weigl
2020/21	Pedraão	Bryan Róchez	Angel Gomes	Naoufel Khacef
2021/22	Abdul Mumin	André Luis	Hélder Ferreira	Gaius Makouta
2022/23	Paulo Eduardo	Emmanuel Boateng	Nicolás Gaitán	João Carvalho
2023/24	Gonçalo Inácio	Ronie Carrillo	Nuno Gonçalo Moreira	Tomás Händel
2024/25	Zé Pedro Figueiredo	Dudu	Jordan Holsgrove	Miguel Menino

Note. The medoid is the player-season that is most representative of an archetype in a season; it is an actual player.

The medoids support the interpretation. Defenders are represented by players with the defender label in all eight seasons. The striker medoids are forwards in six seasons and forwards with a midfielder role in two, and those of the wide attackers combine the forward and midfielder labels in seven seasons; the exception is the 2024/25 medoid, who has only the midfielder label, which indicates that this archetype is the least sharply defined. The central-midfielder medoids have the midfielder label in seven seasons. The medoids are not used in the alignment of the archetypes across seasons (Section 5.2), so their consistency is an independent check of it. Figure 6.1 shows that the composition of the league changes little over time: defenders account for 30 to 35% of the average membership in every season, and the noise cluster is largest in 2024/25 (about 11%).

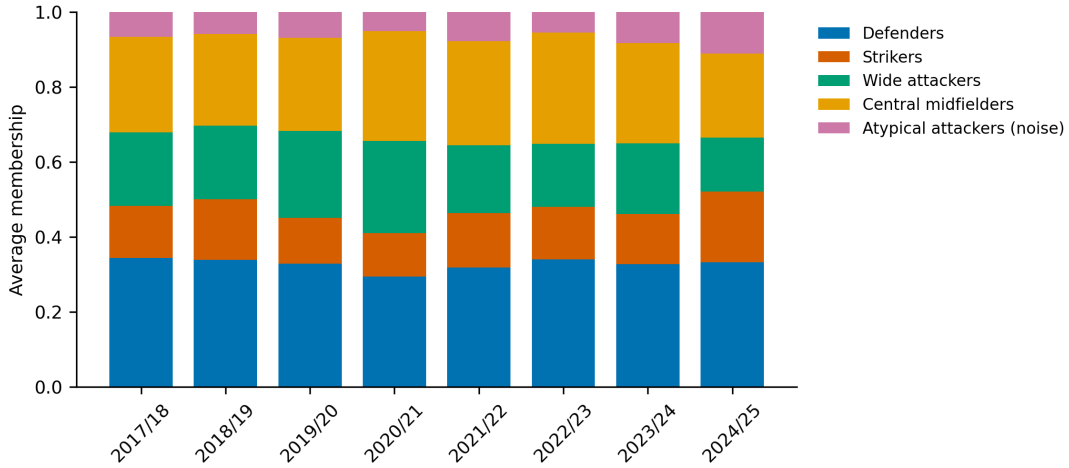


Figure 6.1: Average membership in each archetype by season

6.3 Transitions between archetypes

There are 1,433 pairs of consecutive seasons in which a player has at least 200 minutes in both. Figure 6.2 shows where players go, classified by their largest membership. Defenders are the most persistent (88% remain), followed by central midfielders (69%) and strikers (65%). Wide attackers are the least persistent of the four substantive archetypes: 54% remain, 20% move

to central midfielder and 17% to striker, and 17% of the strikers move to wide attacker. No player moves between defenders and strikers in either direction, and the movements between the attacking archetypes and the noise cluster are frequent: a third of the players in the noise cluster move to wide attackers and a quarter to strikers. The overall persistence is 75% (among players with a substantive archetype in both seasons), and the mean size of the shift in the membership vector is 0.36 (Figure 6.3). The distribution is right-skewed: there are many small shifts and a tail of up to 1.4.

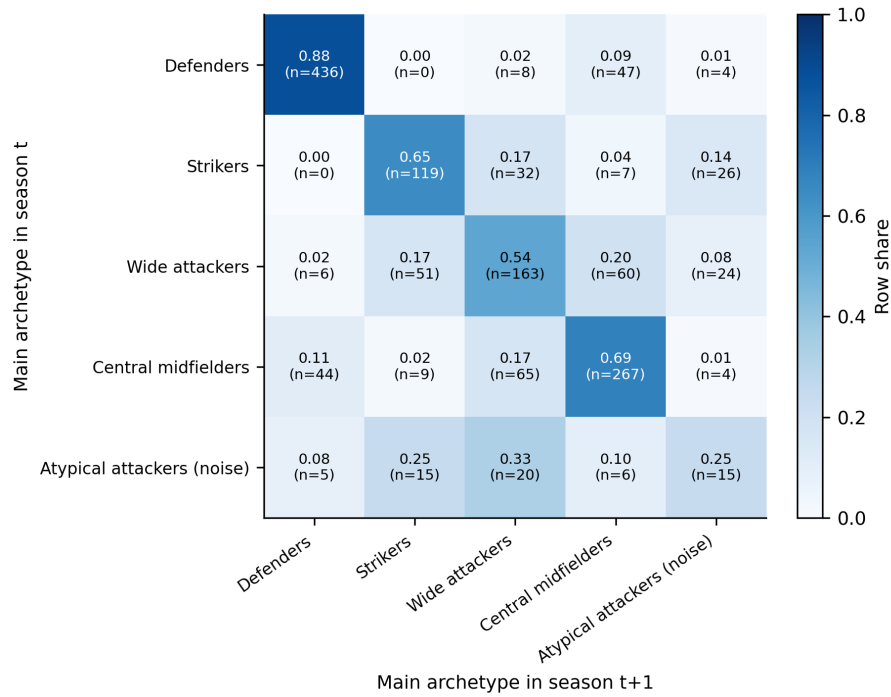


Figure 6.2: Transitions between archetypes (row shares and counts)

Note. Rows give the archetype of largest membership in season t , columns that in season $t + 1$; entries are shares of the row and, in brackets, numbers of players. Based on 1,433 transitions.

Figure 6.4 illustrates the memberships of six players over time. Some players are stable: the memberships of Gonalo In cio are almost entirely in the defender archetype in all five seasons, and Jo o Palhinha is a central midfielder in all four seasons shown, with a temporary defender component in 2018/19 and a wide-attacker component in 2021/22. Others move: Luis D  az is mainly a striker with a wide-attacker component in 2019/20, mainly a wide attacker in 2020/21, and again a striker with a substantial noise component in 2021/22, and Ot  vio moves from central midfielder to wide attacker from 2020/21. Rafa Silva and Paulinho, attacking players with high shot volumes, move among the wide attacker, striker and noise components from one season to the next. Two readings are possible for such movements, a genuine change of role or of form and noise in the estimated memberships, and the data cannot separate them; this is the main reason why the valuation stage has to account for estimation uncertainty.

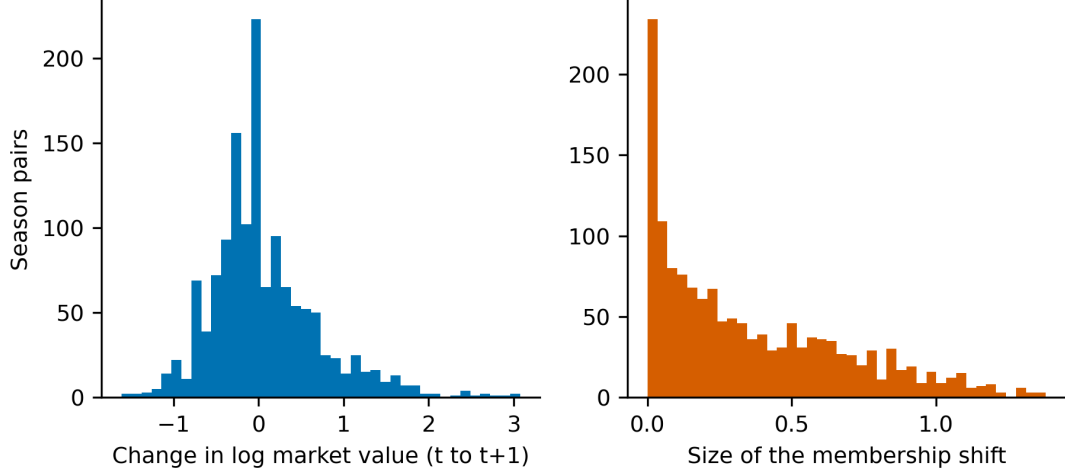


Figure 6.3: Distribution of the change in log market value and of the size of the membership shift

Note. Left: change in log market value between the end of season t and $t + 1$ for the 1,364 pairs with values at both ends. Right: Euclidean norm of the change in the membership vector over the four archetypes, for the 1,433 transitions.

6.4 Transitions and changes in market value

6.4.1 The baseline

The estimation sample contains 1,361 season pairs of 697 players: the 1,364 pairs with a value at both ends, less three for which the team-level control is missing. The core baseline explains 45% of the variance of the change in log value in sample (adjusted R^2 of 0.448) and 44% out of sample in cross-validation with folds that keep a player's pairs together (0.436). This is considerable for a model of changes. It rests on standard determinants only: age, minutes, position, team form, a change of club, and goals and assists. The coefficients (Appendix C, Table C.1) have the expected signs. Value growth falls with age (a coefficient of -0.16 per year, with a small positive curvature), it rises with playing time, both in level and in change (0.19 and 0.37 per unit of log minutes), with the change in the points per match of the team (0.29), and with goals per 90 minutes in level and in change (0.069 and 0.061 per average absolute deviation of the statistic, which is 0.11 goals per 90 minutes). The effects of assists per 90 minutes ($p = 0.06$ for the change), of the position indicators, of the level of team form and of a change of club within the league are not significant. The ten statistics that the full baseline adds do not raise the adjusted R^2 (0.448) and lower the out-of-sample R^2 to 0.422, so goals and assists per 90 minutes carry the performance information that is relevant for value changes in this sample, and the full baseline is mainly a tool to make the test demanding.

6.4.2 Main result

Table 6.4 reports the extended model with the membership changes. The reference archetype is the defenders, which have the largest average membership. All four coefficients are negative: a shift of membership from defenders to any other archetype or to the noise cluster is associated with a smaller increase in value. The coefficient of the wide attackers stands out: -0.22 against

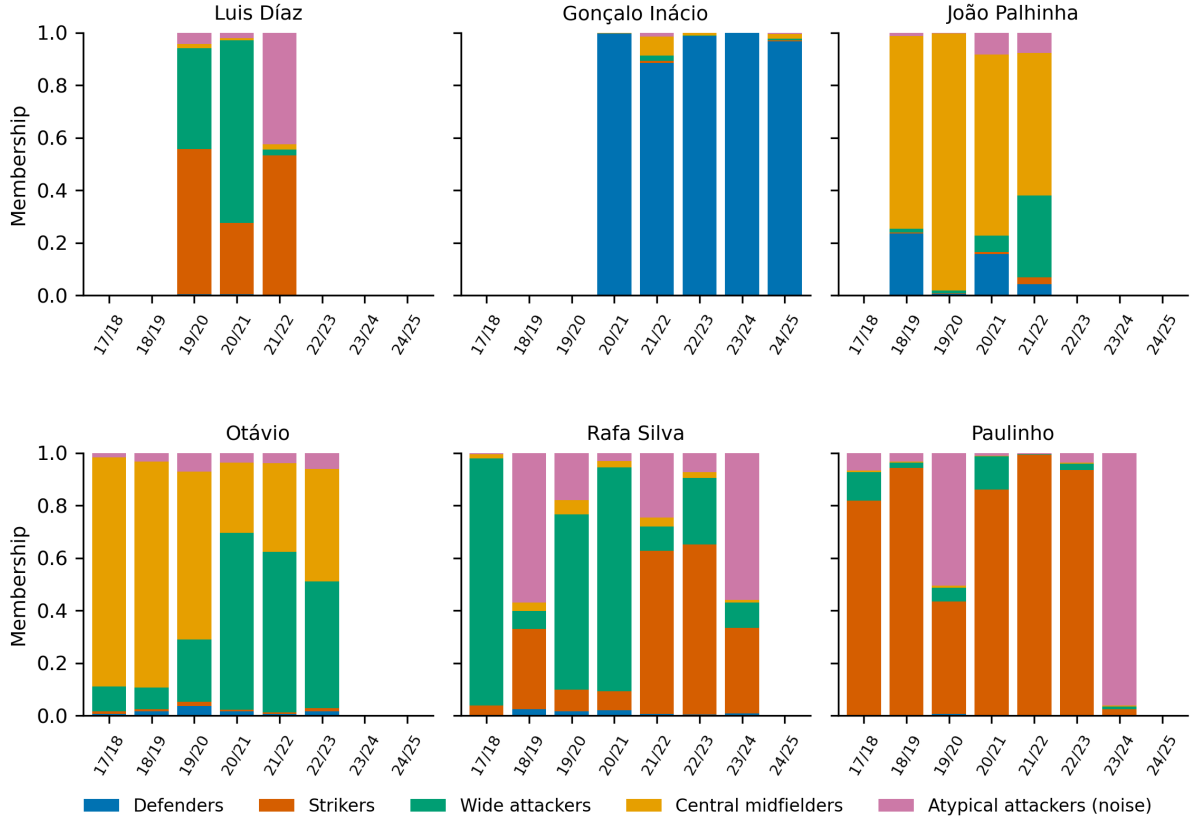


Figure 6.4: Memberships of selected players by season

Note. The players were chosen for illustration. Empty slots are seasons in which the player is not in the sample (fewer than 200 minutes, or not in the Primeira Liga).

the core baseline and -0.31 against the full baseline, which corresponds to a difference in value of about -2.1 and -3.0 per cent for a shift of 0.10 in membership from defenders to wide attackers, and it is significant with bootstrap standard errors ($p = 0.014$ and $p = 0.002$). The coefficients of the strikers (-0.04 and -0.11), of the central midfielders (-0.10 and -0.14) and of the noise cluster (-0.14 and -0.24) are not individually significant.

The joint test is significant against the full baseline and borderline against the core baseline. The Wald test of the four coefficients with the bootstrap covariance matrix gives $p = 0.015$ against the full baseline and $p = 0.079$ against the core baseline; with cluster-robust standard errors that treat the memberships as observed, the values are 0.002 and 0.027 . Table 6.5 adds the gains in explanatory power. Adding the membership changes raises the adjusted R^2 by 0.34 percentage points against the core baseline and by 0.56 points against the full baseline, and the cross-validated out-of-sample R^2 by 0.08 and 0.33 points, with an improvement in 73% and 93% of the cross-validation repetitions. These gains are small. The size of the shift and the change in entropy, which summarise the direction-free movement and the change in specialisation, are not significant, and the levels of the memberships at the start of the period do not help either: it is the direction of the movement, and not its size or the starting point, that is associated with the change in value. A linear summary of the same statistics does not reproduce the association: the first four principal components of the twelve statistics are jointly insignificant against the core baseline ($p = 0.48$; Chapter 7).

Table 6.4: Membership changes and the change in log market value: main configuration

Change in membership of	Core baseline			Full baseline		
	Coef.	Boot. SE	p	Coef.	Boot. SE	p
Strikers	-0.036	0.107	0.738	-0.114	0.125	0.363
Wide attackers	-0.216	0.088	0.014	-0.308	0.101	0.002
Central midfielders	-0.104	0.079	0.188	-0.140	0.087	0.108
Noise (atypical attackers)	-0.141	0.147	0.340	-0.239	0.183	0.192

Note. Dependent variable: change in log market value from the end of season t to $t+1$. Reference archetype: defenders. Each coefficient is the association of a full shift of membership from defenders to the archetype named, with the controls of the baseline held fixed (see Section 5.4); a shift of 0.10 corresponds to one tenth of the coefficient. Bootstrap standard errors and p -values are based on 500 replications of both stages (Section 5.5). The Wald test of the four coefficients gives $p = 0.079$ (core) and $p = 0.015$ (full). $N = 1,361$ pairs of 697 players.

Table 6.5: Explanatory power of alternative summaries of the membership dynamics: main configuration

Added to the baseline (number of variables)	Core baseline			Full baseline		
	p	$\Delta \bar{R}^2$	ΔR_{oos}^2	p	$\Delta \bar{R}^2$	ΔR_{oos}^2
Membership changes Δu (4 variables)	0.027	+0.34	+0.08	0.002	+0.56	+0.33
Size of the shift (1)	0.396	-0.01	-0.07	0.395	-0.01	-0.07
Change in entropy (1)	0.981	-0.04	-0.09	0.513	-0.02	-0.07
Membership levels u_t (4)	0.863	-0.11	-0.33	0.847	-0.10	-0.33
Levels and changes (8)	0.080	+0.32	-0.14	0.009	+0.52	+0.09

Note. p is the cluster-robust Wald p -value of the added block, treating the memberships as observed. $\Delta \bar{R}^2$ is the change in adjusted R^2 and ΔR_{oos}^2 the change in cross-validated out-of-sample R^2 (five folds grouped by player, 30 repetitions), both in percentage points. $N = 1,361$.

6.4.3 Player effects

Player-specific characteristics that are not in the baseline could be correlated with the memberships. The Hausman test rejects the random-effects model in favour of fixed effects for both baselines ($p < 0.001$), as expected, since talent and reputation are likely to be correlated with the profile of a player. The fixed-effects estimates rest on the players with at least two season pairs, a subsample of about 350 players. In this model the membership changes remain jointly significant: $p = 0.018$ against the core baseline and $p = 0.001$ against the full baseline. Age and its square are not identified together with player and season effects and are left out of these estimates (Section 5.4).

6.5 Summary

In the main configuration, the clustering stage yields four interpretable archetypes of outfield players (defenders, central midfielders, wide attackers and strikers) and a small noise cluster of high-volume attackers. Archetypes are reasonably persistent across seasons (75% of players keep their archetype), but the memberships of attacking players are volatile. In the valuation stage, a model with age, playing time, position, team form, club change and goals and assists explains 45% of the variance of value changes. The membership changes are jointly significant against the full baseline (bootstrap $p = 0.015$) and borderline against the core baseline ($p = 0.079$), and they

add 0.3 to 0.6 percentage points to the adjusted R^2 and 0.1 to 0.3 points to the out-of-sample R^2 . The association is concentrated in the shift from the defender profile to the wide-attacker profile and is present with player fixed effects; it is not present for the size of the shift or the change in entropy. This is the result of one run of a stochastic clustering, with one setting of the tuning parameters. The next chapter shows how far it carries over to other settings and to other random draws.

Chapter 7

Robustness and Sensitivity

This chapter examines how far the results of Chapter 6 depend on the choices made in the analysis and on chance. It starts with the inference (Section 7.1), turns to the sample and the outcome (Section 7.2), to the way in which the transitions are summarised and to the baseline (Section 7.3), and to the number of archetypes and the rate variables (Section 7.4). Section 7.5 turns to the fuzziness of the memberships and the noise multiplier, Section 7.6 to the alignment of the archetypes, and Section 7.7 to the random draw of the clustering. Because many specifications are examined, p -values are not corrected unless stated, and the pattern across specifications is more informative than any single value. The main configuration of Chapter 6 remains the reference throughout.

7.1 Estimated memberships: the bootstrap

Table 7.1 compares the conventional Wald tests, which treat the memberships as observed, with the tests based on the two-stage bootstrap (Section 5.5) for the main fuzziness and noise settings ($m = 1.3$, $\rho = 1.0$) and different numbers of archetypes and rate variables. The bootstrap standard errors are larger than the conventional ones by 7 to 46% depending on the cell and the baseline, 19% on average, and the correction is largest where the memberships are least reliable: for $K = 3$ with shrunk rates it raises the p -value of the core test from 0.27 to 0.56 and that of the full test from 0.05 to 0.34. For the main configuration the ratio is 1.08 and 1.15, and the joint p -values change from 0.027 to 0.079 and from 0.002 to 0.015, so the correction matters for the conclusion: against the core baseline the association is no longer significant at 5%. The mean of the bootstrap estimates also differs from the original coefficients, by up to 0.17 for individual coefficients in the settings reported below and in the direction of zero in the cases with the largest differences, as would be expected if noise in the memberships attenuated the coefficients. The bootstrap corrects the standard errors but not this attenuation, so the point estimates should be read as conservative.

7.2 Sample and outcome

The association in the main configuration is not driven by the definition of the sample or the outcome (Table 7.2). Against the full baseline, the block of membership changes is significant at

Table 7.1: Conventional and bootstrap tests by number of archetypes and rate variables ($m = 1.3$, $\rho = 1.0$)

Rates	K	Core: p		Full: p		SE ratio	Full baseline	
		Conv.	Boot.	Conv.	Boot.		$\Delta \bar{R}^2$	ΔR_{oos}^2
Shrunk	3	0.266	0.557	0.053	0.335	1.40	+0.17	+0.01
Shrunk	4	0.027	0.079	0.002	0.015	1.11	+0.56	+0.33
Shrunk	5	0.428	0.723	0.522	0.757	1.30	-0.02	-0.25
Raw	3	0.054	0.077	0.003	0.012	1.10	+0.58	+0.47
Raw	4	0.034	0.055	0.002	0.016	1.17	+0.59	+0.40

Note. Wald p -values of the block of membership changes with cluster-robust (conventional) and bootstrap covariance matrices (500 replications). SE ratio is the average over both baselines of the mean ratio of the bootstrap to the conventional standard errors. $\Delta \bar{R}^2$ and ΔR_{oos}^2 are the gains in adjusted and cross-validated R^2 against the full baseline, in percentage points. The row for shrunk rates and $K = 4$ is the main configuration.

5% in seven of the eight variations, and against the core baseline in five of eight, with changes in the adjusted R^2 that are positive in all of them (0.09 to 0.39 percentage points for the core baseline and 0.23 to 0.56 for the full baseline). The exceptions are the sample without the pandemic seasons (960 pairs), against both baselines, the players who did not change club (1,058 pairs) against the core baseline ($p = 0.24$; the full-baseline test is just significant at $p = 0.046$), and the players aged up to 30 against the core baseline ($p = 0.064$): in these the gains stay positive but the tests lose power. The result is the same with player fixed effects (Section 6.4). These variations and the fixed-effects comparison were computed for the main configuration only, with conventional standard errors.

Table 7.2: Sample and outcome variations (main configuration)

Variation	Pairs	Players	Core baseline		Full baseline	
			p	$\Delta \bar{R}^2$	p	$\Delta \bar{R}^2$
All pairs (main configuration)	1,361	697	0.027	+0.34	0.002	+0.56
Winsorised outcome (1st and 99th percentile)	1,361	697	0.033	+0.30	0.004	+0.50
Players who did not change club	1,058	611	0.240	+0.09	0.046	+0.23
Non-zero changes in value only	1,164	632	0.047	+0.31	0.010	+0.46
At least 500 minutes in both seasons	1,069	574	0.023	+0.39	0.015	+0.44
Without pairs starting in 2019/20 and 2020/21	960	598	0.270	+0.16	0.158	+0.23
Players aged 30 or younger	1,222	642	0.064	+0.28	0.007	+0.52
Without noise-dominated players (noise membership > 0.5)	1,283	669	0.035	+0.33	0.009	+0.49

Note. Cluster-robust Wald p -values of the block of membership changes; $\Delta \bar{R}^2$ is the change in adjusted R^2 in percentage points.

7.3 Summaries of the transitions and the baseline

Direction, size and specialisation. Replacing the membership changes by the size of the shift or by the change in entropy removes the association in the main configuration (Table 6.5), and so does adding the membership levels at the start of the season. What is associated with the change in value is the direction of the movement.

Linear summaries. Principal component scores condense the same twelve statistics linearly. The first four components explain about 71% of their variance. Their changes are jointly insignificant against the core baseline ($p = 0.48$, change in adjusted R^2 of -0.01 percentage points), and so are those of the first eight ($p = 0.59$). They cannot be tested against the full baseline, which already contains the changes of every statistic. This benchmark does not depend on m or on the random draw. Because the membership changes are significant at several values of m (Section 7.5) where the linear summary is not, any association that the memberships carry is not a reflection of linear changes in the performance statistics.

Controls without performance statistics. The overlap between the memberships and the performance controls is the reason for the full baseline, and the opposite check removes the performance controls altogether. A baseline with only age, playing time, position, team form, club change and season effects has an adjusted R^2 of 0.433, compared with 0.448 once goals and assists are added. Table 7.3 shows that the membership changes do not behave differently against this baseline: they are significant at $m = 1.3$ ($p = 0.020$) and more clearly at $m = 1.4$ and $m = 1.5$ ($p = 0.001$ and $p < 0.001$ with conventional standard errors), with gains in out-of-sample R^2 of 0.44 and 0.59 percentage points at these two values. The membership changes are therefore not proxies for goals and assists.

Table 7.3: Benchmarks for the membership changes

<i>A. Principal component scores against the core baseline</i>			
	p	$\Delta \bar{R}^2$	ΔR_{os}^2
Principal components, first 4	0.483	-0.01	—
Principal components, first 8	0.589	+0.00	—
<i>B. Membership changes against the baseline without performance statistics</i>			
Fuzziness m	p	$\Delta \bar{R}^2$	ΔR_{os}^2
1.2	0.108	+0.26	-0.04
1.3	0.020	+0.39	+0.13
1.4	0.001	+0.63	+0.44
1.5	0.000	+0.74	+0.59
2.0	0.005	+0.53	+0.36

Note. Cluster-robust Wald p -values, treating the memberships as observed; changes in adjusted and cross-validated R^2 in percentage points. Panel B: shrunk rates, $K = 4$, $\rho = 1.0$. $N = 1,361$.

7.4 Number of archetypes and rate variables

At $m = 1.3$ the number of archetypes and the treatment of the rates matter more than the sample (Table 7.1). With the bootstrap, three of the ten tests are below 5% (shrunk rates with $K = 4$, raw rates with $K = 3$ and raw rates with $K = 4$, all against the full baseline), three more lie between 5 and 8% (all against the core baseline), and the other four are well above. With shrunk rates, $K = 3$ and $K = 5$ show no association (bootstrap p -values of 0.34 and 0.76 against the full baseline); with raw rates, $K = 3$ and $K = 4$ do. The gains are small: where the full-baseline test is significant, the adjusted R^2 rises by 0.56 to 0.59 percentage points and the out-of-sample R^2 by 0.33 to 0.47 points. The association is therefore not specific to the shrunk

rates, which also appears in the specification grid below, but it is specific to some numbers of archetypes, and $K = 5$ stands out as the setting in which it is most often absent (Table 7.5).

7.5 Fuzziness and the noise multiplier

The fuzziness exponent m is the setting on which the results depend most. Table 7.4 fixes the other settings of the main configuration and varies m from 1.2 to 2.0. Larger values of m spread the memberships more evenly, as the column on crispness shows: the share of players with a largest membership of at least 0.7 falls from 76% at $m = 1.2$ to 63% at $m = 1.3$ (the main configuration), 49% at $m = 1.4$, 37% at $m = 1.5$ and 4% at $m = 2.0$, where the memberships are almost uniform and hardly describe archetypes. The noise share rises from 4.8% to 12%, and the persistence of archetypes is between 0.67 and 0.75 for m from 1.2 to 1.5 and falls to 0.54 at $m = 2.0$.

Table 7.4: Sensitivity to the fuzziness exponent m (shrunk rates, $K = 4$, $\rho = 1.0$)

m	Crisp	Noise	Persist.	XB	Core baseline			Full baseline		
					p	$\Delta\bar{R}^2$	ΔR_{oos}^2	p	$\Delta\bar{R}^2$	ΔR_{oos}^2
1.2	76%	4.8%	0.71	0.89	0.445	+0.09	-0.22	0.277	+0.18	-0.07
1.3	63%	5.3%	0.75	0.72	0.079	+0.34	+0.13	0.015	+0.56	+0.43
1.4	49%	6.3%	0.70	0.66	0.037	+0.42	+0.28	0.008	+0.66	+0.58
1.5	37%	7.3%	0.67	0.75	0.029	+0.52	+0.41	0.008	+0.75	+0.72
2.0	4%	12.0%	0.54	0.63	0.088	+0.22	+0.06	0.019	+0.43	+0.25

Note. Crisp is the share of players whose largest substantive membership is at least 0.7; Noise the share assigned to the noise cluster; Persist. the share who keep their archetype; XB the Xie–Beni index averaged over seasons (lower is better). p is the bootstrap Wald p -value of the block of membership changes (500 replications); $\Delta\bar{R}^2$ and ΔR_{oos}^2 are gains in adjusted and cross-validated R^2 in percentage points (ten cross-validation repetitions, so the row $m = 1.3$ differs slightly from Table 6.5, which uses 30). The row $m = 1.3$ is the main configuration.

With near-crisp memberships ($m = 1.2$) there is no association after the bootstrap ($p = 0.45$ and 0.28). From $m = 1.3$ on the membership changes are jointly significant against the full baseline ($p = 0.015$, 0.008, 0.008 and 0.019 at $m = 1.3$, 1.4, 1.5 and 2.0), and against the core baseline they are significant at $m = 1.4$ and $m = 1.5$ ($p = 0.037$ and 0.029) and borderline at $m = 1.3$ and $m = 2.0$ ($p = 0.079$ and 0.088). The adjusted R^2 rises by 0.4 to 0.5 percentage points against the core baseline and by 0.7 to 0.8 points against the full baseline at $m = 1.4$ and 1.5, and the out-of-sample R^2 by 0.3 to 0.4 and by 0.6 to 0.7 points. The values $m = 1.4$ and $m = 1.5$ lie within the range of 1 to 1.5 that is recommended for medoid-based methods, and $m = 1.5$ is one of the three values compared by D’Urso et al. (2022). Within the main configuration the Xie–Beni index is lowest at $m = 1.4$ among the values from 1.2 to 1.5, which is the value that the proposal’s selection rule would have chosen; this was not the rule used (Chapter 5), but the reader should know that the results at that value are somewhat stronger than in the main configuration.

The archetypes are the same types at $m = 1.3$, 1.4 and 1.5. The medoids are defenders in all eight seasons for one archetype, forwards for the strikers, forwards with a midfield role for the wide attackers, and midfielders for the central midfielders. The numbering of the archetypes differs between runs, because it is inherited from the first season, so coefficients are compared

by the type of archetype. In all three runs the reference is the defenders and all coefficients are negative, and the one that stands out is that of the wide attackers: -0.22 and -0.31 against the core and full baselines at $m = 1.3$, -0.26 and -0.36 at $m = 1.4$ (bootstrap $p = 0.004$ and 0.001), and -0.31 and -0.41 at $m = 1.5$ ($p = 0.003$ and 0.001). The shift from the defender profile to the wide-attacker profile is therefore the association that is found, and it grows in size with the fuzziness. A shift of 0.10 corresponds to a difference in value of about -2 to -3 per cent at $m = 1.3$ and about -3 to -4 per cent at $m = 1.5$.

Noise multiplier. A lower noise multiplier ($\rho = 0.75$) increases the noise share to 14% and a higher one ($\rho = 1.5$) lowers it to 1.5% . At $m = 1.3$ the conventional p -values are 0.046 (core) and 0.002 (full) at $\rho = 0.75$, 0.027 and 0.002 at the main value $\rho = 1.0$, and 0.099 and 0.014 at $\rho = 1.5$; with the bootstrap, $\rho = 1.5$ gives 0.171 and 0.069 . The association is therefore weaker when fewer players are put into the noise cluster.

The whole specification space. Table 7.5 summarises the 120 combinations of rates, number of archetypes, noise multiplier and fuzziness, and Figure 7.1 shows the conventional p -values. Against the core baseline, 34 of the 120 specifications (28%) have a p -value below 5%; against the full baseline 74 (62%) do. The adjusted R^2 increases in 92 (core) and 107 (full) of the specifications, by a median of 0.13 and 0.28 percentage points and by at most 0.83 points. The out-of-sample R^2 increases in 43 (core) and 79 (full) of the 120 specifications, with medians of -0.08 and $+0.13$ points and a maximum of 0.72 points. The proportion of significant results is highest at $K = 4$ (53% core, 83% full) and lowest at $K = 5$ (10% and 30%), highest at $m = 1.5$ (38% core, 83% full) and, against the full baseline, lowest at $m = 1.2$ (42%) and $m = 2.0$ (50%). It is the same for shrunk and raw rates against the core baseline (28% each) and somewhat higher for raw rates against the full baseline (70% against 53%). The map in Figure 7.1 is patchy and not a smooth function of m and K , which suggests that part of the pattern reflects the idiosyncrasies of individual clustering solutions and of the random draw (Section 7.7).

Multiple comparisons. The bootstrap was applied to twenty comparisons (ten configurations against two baselines). The smallest bootstrap p -value is 0.0076 . Neither if the ten comparisons of the fuzziness grid are treated as one family nor if all twenty are does any comparison survive a Holm correction at the 5% level, since the thresholds are 0.005 and 0.0025 . The specifications are not independent tests of different hypotheses, which makes this correction conservative, but it shows that the evidence is not strong enough to rely on any single cell.

7.6 Consistency of the archetype labels across seasons

The alignment of the archetypes across seasons determines whether a change in a membership is a real transition. The first version of the analysis matched archetypes by the distance between medoids. Its quality was checked for every one of the 120 clustering runs of the specification grid with information that the matching does not use: the profiles of the archetypes. For each pair of adjacent seasons, the check asks whether label k in season t is matched to label k in season $t + 1$ when the profiles are matched optimally; a pair is inconsistent if not.

Table 7.5: Summary of the specification grid (120 specifications)

	N	Share with $p < 0.05$		Median $\Delta \bar{R}^2$		Median ΔR_{os}^2	
		Core	Full	Core	Full	Core	Full
All specifications	120	28%	62%	+0.13	+0.28	-0.08	+0.13
<i>By fuzziness m</i>							
$m = 1.2$	24	25%	42%	+0.07	+0.17	-0.15	-0.04
$m = 1.3$	24	17%	67%	+0.16	+0.37	-0.09	+0.17
$m = 1.4$	24	38%	67%	+0.13	+0.35	-0.07	+0.20
$m = 1.5$	24	38%	83%	+0.18	+0.47	-0.07	+0.20
$m = 2.0$	24	25%	50%	+0.10	+0.19	-0.08	+0.02
<i>By number of archetypes K</i>							
$K = 3$	30	23%	67%	+0.12	+0.26	-0.01	+0.17
$K = 4$	30	53%	83%	+0.20	+0.40	+0.04	+0.26
$K = 5$	30	10%	30%	+0.01	+0.12	-0.17	-0.09
$K = 6$	30	27%	67%	+0.17	+0.37	-0.10	+0.10
<i>By noise multiplier ρ</i>							
$\rho = 0.75$	40	28%	70%	+0.14	+0.31	-0.06	+0.18
$\rho = 1.0$	40	22%	50%	+0.07	+0.22	-0.10	+0.06
$\rho = 1.5$	40	35%	65%	+0.13	+0.27	-0.07	+0.10
<i>By rate variables</i>							
Shrunk	60	28%	53%	+0.13	+0.23	-0.08	+0.08
Raw	60	28%	70%	+0.13	+0.34	-0.09	+0.16

Note. Specifications combine shrunk or raw rates, $K \in \{3, 4, 5, 6\}$, $\rho \in \{0.75, 1.0, 1.5\}$ and $m \in \{1.2, 1.3, 1.4, 1.5, 2.0\}$. p -values are conventional cluster-robust Wald tests and treat the memberships as observed. Gains in adjusted and cross-validated R^2 (ten repetitions) are in percentage points. N is the number of specifications in the row.

The check found consistent labels in every season pair in only 36 of the 120 runs (30%). The share falls steeply with the number of clusters: 73% for $K = 3$, 27% for $K = 4$, 17% for $K = 5$ and 3% for $K = 6$. For the shrunk-rate runs with $K = 4$ and $\rho = 1.0$ it was consistent at $m = 1.3$ (the main configuration) and $m = 1.4$, and inconsistent in one season pair at $m = 1.5$ (2022/23 to 2023/24), in two at $m = 2.0$ and in three at $m = 1.2$. The inconsistency at $m = 1.5$ agrees with what the medoids showed: the central-midfielder and striker medoids of 2023/24 and 2024/25 appeared under the numbers that held other archetypes earlier. For this reason the alignment uses the profiles of the archetypes (Section 5.2), and *all results in Chapters 6 and 7, including the specification grid and the bootstrap, were recomputed with this alignment*. This is a correction of the method made because of the label check and not because of any outcome. After the correction, the labels are consistent in all 120 runs by construction; the independent check is the position labels of the medoids, which do not enter the alignment. The share of seasons in which an archetype's medoid has the archetype's most common position label is 0.88 at $m = 1.2$ and 1.3, 0.84 at $m = 1.4$, 0.75 at $m = 1.5$ and 0.59 at $m = 2.0$ in the $K = 4$ runs with shrunk rates, and between 0.74 and 0.79 on average for each K over all runs. The archetypes are thus well defined at $m \leq 1.4$ and less so at $m = 1.5$ and $m = 2.0$, where the memberships are softer.

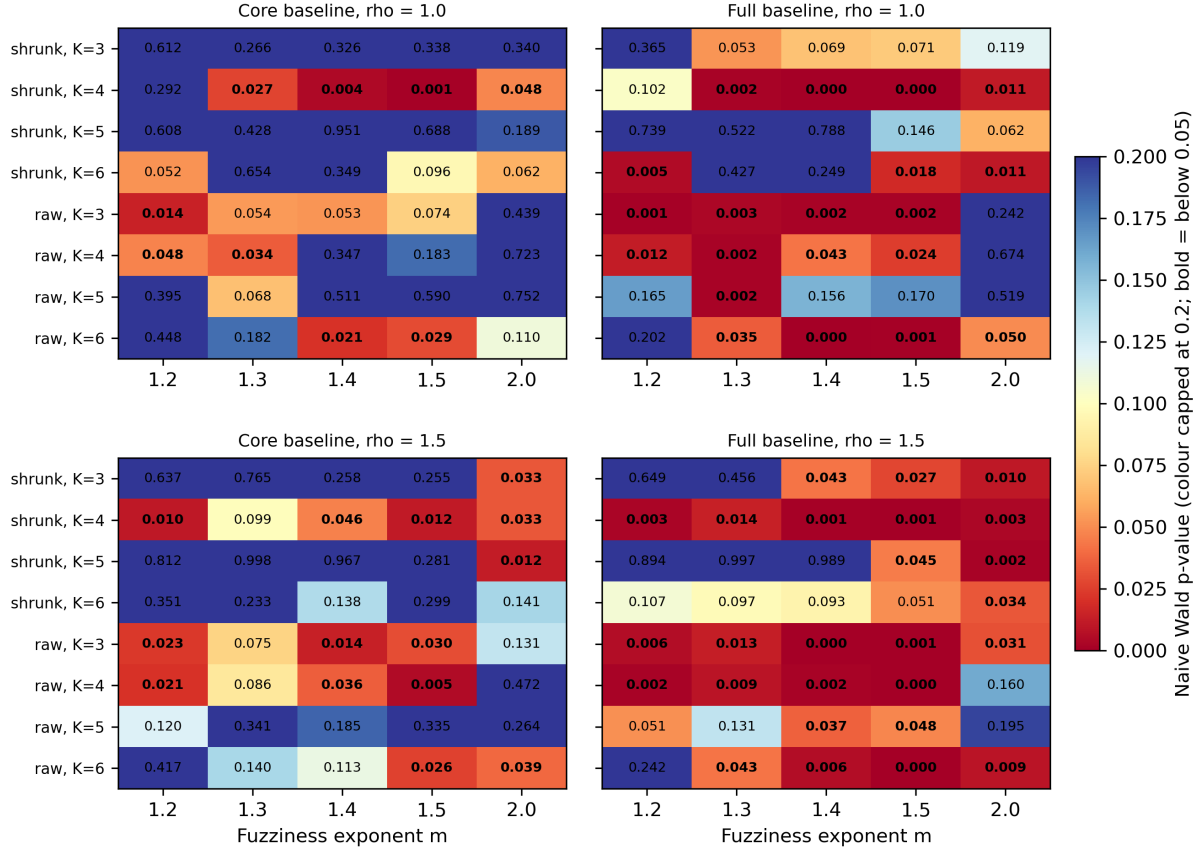


Figure 7.1: Conventional Wald p -values across the specification grid

Note. Each cell is one specification. Colour is capped at $p = 0.2$; bold values are below 0.05. The left panels use the core baseline, the right panels the full baseline; the upper panels $\rho = 1.0$ and the lower panels $\rho = 1.5$.

7.7 The random draw of the clustering

The clustering starts from random medoids, and the partitions of different starts agree only moderately (an adjusted Rand index of about 0.5, Table 6.1). The best of ten starts is kept for every season, but the objective is flat, so that another seed, or another ordering of the rows, can yield different memberships and different test results. This was seen directly: a run of the same main configuration made before the correction of the alignment and of the data (the data correction is described in Chapter 4), with a different ordering of the rows, gave conventional p -values of 0.70 (core) and 0.36 (full), against 0.027 and 0.002 in the run reported in Chapter 6. To measure the dependence on the draw, the first stage (the clustering of all seasons and the alignment) was repeated with 30 different seeds for each fuzziness value, and the key test was repeated each time (Table 7.6, Figure 7.2).

The p -values vary by orders of magnitude between draws: at $m = 1.3$ the full-baseline p -value ranges from below 0.001 to 0.32 and the core-baseline value from 0.003 to 0.71. Against the full baseline, the test is significant in 60% of the draws at $m = 1.3$ (median $p = 0.025$), 67% at $m = 1.4$ (0.008) and 73% at $m = 1.5$ (0.016), and in 40% and 43% at $m = 1.2$ and $m = 2.0$. Against the core baseline it is significant in 37%, 37% and 27% of the draws at $m = 1.3$, 1.4 and 1.5 (medians of 0.08 to 0.11). The median gain in adjusted R^2 is 0.26 to 0.38 percentage points

Table 7.6: Results over 30 random draws of the clustering (shrunk rates, $K = 4$, $\rho = 1.0$)

m	Core baseline			Full baseline				Median $\Delta \bar{R}^2$		Median ΔR_{os}^2	
	Median p	$p < 0.05$	Median p	IQR	Max	$p < 0.05$		Core	Full	Core	Full
1.2	0.224	17%	0.086	0.008–0.145	0.50	40%		+0.05	+0.22	-0.15	+0.03
1.3	0.113	37%	0.025	0.004–0.083	0.32	60%		+0.13	+0.26	-0.07	+0.12
1.4	0.082	37%	0.008	0.002–0.070	0.31	67%		+0.14	+0.38	-0.02	+0.23
1.5	0.101	27%	0.016	0.005–0.051	0.55	73%		+0.10	+0.33	-0.07	+0.17
2.0	0.266	17%	0.093	0.015–0.192	0.88	43%		+0.04	+0.15	-0.11	+0.04

Note. Conventional cluster-robust Wald p -values of the block of membership changes; $p < 0.05$ is the share of draws in which the test is significant; IQR is the interquartile range of the p -values; gains in adjusted and cross-validated R^2 (ten repetitions) are medians in percentage points.

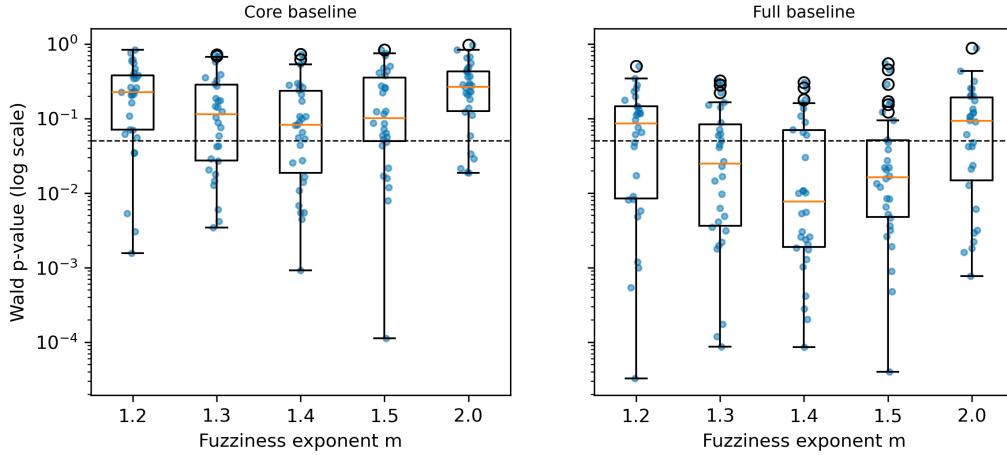


Figure 7.2: Wald p -values over 30 random draws of the clustering, by fuzziness

Note. Each point is one draw; boxes show the quartiles; the dashed line is $p = 0.05$; the scale is logarithmic.

against the full baseline and 0.10 to 0.14 points against the core baseline. Out of sample, the median gain is positive against the full baseline (0.12 to 0.23 points, with a positive gain in 70 to 77% of the draws) but negative against the core baseline (between -0.02 and -0.07 points, positive in only 40 to 43% of the draws). The run reported in Chapter 6 is a favourable draw: its conventional p -values (0.027 and 0.002) are lower than the medians over draws (0.11 and 0.025) and its out-of-sample gains are larger. A typical draw therefore shows an association against the full baseline, but not an improvement in out-of-sample prediction over the core baseline, which is the better predictor (cross-validated R^2 of 0.436 against 0.422 for the full baseline). Against the full baseline, which overfits, the membership changes improve prediction in most draws; this is consistent with, but does not show, that they act partly as a regulariser for the twenty-four additional statistics.

7.8 Estimated attribute weights

The estimation of the attribute weights, which the proposal envisaged, was tried first and abandoned (Section 5.1). Appendix B reports the weights obtained for the 2023/24 season in the variants that were tried. They show that the failure is systematic and not specific to one setting:

in the variants that avoid collapse onto the success rates, positions take most of the weight, and with five clusters the weights collapse onto the positions in all four variants.

7.9 Summary

The association between membership changes and changes in market value in the main configuration is robust to the definition of the sample and the outcome, to player fixed effects, and to the inclusion or exclusion of performance controls, and it is not reproduced by a linear summary of the same statistics. It depends on the clustering. It is absent for near-crisp memberships ($m = 1.2$), for $K = 5$ and, at $m = 1.3$, for $K = 3$ with shrunk rates, strongest at $m = 1.4$ to 1.5, and weaker when fewer players are assigned to the noise cluster. It varies greatly with the random draw of the clustering, in which it is found against the full baseline in about two thirds of the draws and against the core baseline in about a third, and in which the out-of-sample gain over the core baseline is typically absent. Applying a correction for the number of comparisons leaves no result significant. The evidence is therefore weak and specification-dependent: it points to a small association between the movement from the defender profile to the wide-attacker profile and changes in value, but not to a robust improvement in explanatory power.

Chapter 8

Discussion

This chapter interprets the findings, relates them to the literature, discusses what they mean for practice, and sets out the limitations that the reader should keep in mind.

8.1 What the thesis finds

Archetypes and transitions. The first supporting question asked which archetypes a robust mixed-data fuzzy clustering identifies in the Primeira Liga and how players move between them. Four archetypes of outfield players are identified, defenders, central midfielders, wide attackers and strikers, together with a small noise cluster of high-volume attackers whose output resembles none of them. The archetypes are recognisable from their medoids and statistics, and they persist over eight seasons: the same types of players are the medoids, and 75% of players keep their archetype from one season to the next. Persistence differs greatly by archetype. Defenders are almost fixed (88% remain), while wide attackers (54%), and to a lesser extent strikers (65%) and the noise cluster (25%), are volatile, and movements occur mostly between adjacent roles, from wide attackers to central midfielders or strikers, and never between defenders and strikers. Part of the volatility of attacking players may reflect genuine changes of role or form, and part of it noise in the memberships, which the data cannot separate.

Explanatory power for changes in value. The main question was whether membership transitions have incremental explanatory power for changes in market value beyond individual performance metrics and basic characteristics. In the main configuration the answer is a qualified yes. The membership changes are jointly significant against the full baseline (bootstrap $p = 0.015$) and borderline against the core baseline ($p = 0.079$); they raise the adjusted R^2 by 0.3 to 0.6 percentage points and the out-of-sample R^2 by 0.1 to 0.3 points; the association is present with player fixed effects and under most variations of the sample and the outcome; and a linear summary of the same statistics, the principal components, shows none. The association is concentrated in one movement, from the defender profile to the wide-attacker profile, which is associated with a smaller increase in value, by about 2 to 3 per cent for a shift of 0.10 in membership.

The qualification is considerable. First, the association depends on the clustering. It is absent for near-crisp memberships and for five archetypes, strongest for moderately soft memberships,

and weaker when fewer players are assigned to the noise cluster. Second, it depends on the random draw: in 30 repeated clusterings of the main configuration, the full-baseline test is significant in 60% of the draws and the core-baseline test in 37%, with p -values that range over several orders of magnitude, and the run reported in the Results is more favourable than a typical draw. Third, the improvement in out-of-sample prediction that the memberships bring is found against the full baseline, which predicts worse than the core baseline because of its twenty-four additional statistics, but in a typical draw not against the core baseline itself. Fourth, no result survives a correction for the number of comparisons. The most accurate summary is that there is weak and specification-dependent evidence of a small association between the movement from a defender to a wide-attacker profile and changes in market value, and that the thesis does not provide evidence that membership transitions improve the prediction of changes in value.

8.2 Interpreting the dependence on the clustering

The most informative finding of the sensitivity analysis is how the association depends on the fuzziness exponent. Three observations suggest a reading, which remains a conjecture because it was not tested directly. First, with $m = 1.2$ the memberships are close to crisp, since 76% of players have a largest membership of at least 0.7, so that a transition is mostly either a jump between archetypes or no change. Such a variable discards the graded information on where a player lies between archetypes and, since a small change in the data can move a player over the boundary, is also a noisy one, which attenuates its coefficient; no association is found. Second, at $m = 1.4$ and 1.5 the memberships are graded (37 to 49% have a largest membership of at least 0.7) while the archetypes are still well defined, and changes in them carry more of the continuous movement of a player in the space of statistics; the association is strongest here. Third, at $m = 2.0$ the memberships are almost uniform (4% at least 0.7), they hardly describe archetypes any more, and the association weakens. That the principal component scores of the same statistics show no association supports the view that what the memberships capture is not a linear change in the underlying statistics, but the relative position of a player with respect to the archetypes.

The variation across random draws, in turn, shows that this reading cannot be pressed too far: a different draw at the same m changes the p -value by orders of magnitude. The memberships are estimated with substantial noise, and the part of the association that survives across settings is better described by its median over draws than by any single run.

The substantive content of the association is limited. In every configuration for which coefficients are shown, all coefficients are negative relative to the defender profile, and the one that stands out is that of the wide attackers: -0.22 against the core baseline and -0.31 against the full baseline at $m = 1.3$, rising to -0.31 and -0.41 at $m = 1.5$. A shift of 0.10 in membership is associated with a change in value of about -2 to -4 per cent, to be compared with a standard deviation of the change in log value of 0.63. Several explanations are possible, and the data in this thesis cannot discriminate between them. Valuations may depend on the role in ways that a baseline with a position indicator and a few statistics does not capture, for instance through different age profiles of value for attackers and defenders; a deviation from a stable, defensive profile towards a wide attacking one may signal a less settled role; or a large attacking output

in one season may regress towards the mean in the next, which would link movements towards the wide-attacker profile to subsequent lower valuations. None of these has been examined here.

8.3 Relation to the literature

The thesis extends the work of D’Urso et al. (2022) in the directions that they themselves suggest, by adding the temporal dimension and a financial variable. It also contains a methodological result that is relevant for anyone applying their model. The estimation of the attribute weights, which is the distinguishing feature of the model, breaks down when the groups of variables differ in dimension and in the number of distinct values, as they do in this and in most football data sets, because it favours the most compact group. The weights then had to be fixed in advance, in line with Akhanli and Hennig (2023). The alignment of the archetypes across seasons also needed care: matching them by their medoids, which are single players, gave inconsistent labels in most clustering runs, whereas matching them by their profiles did not. In contrast to Pacifico (2025), who works with a panel of 27 players, the longitudinal analysis here covers all outfield players of a league over eight seasons, with a link to market values, and the finding that transitions are only weakly related to market values is a qualification of the suggestion that monitoring cluster transitions would be informative for practitioners. The baseline results are in line with the valuation literature (Franceschi et al., 2024): value growth falls with age, increases with playing time and with recent form of the team, and increases with goals per 90 minutes, while position and a change of club within the league have no significant association in the baseline. The thesis responds to the call of Franceschi et al. (2024) to study leagues outside the big five, with the Primeira Liga as the example they name, and it confirms the concern of Müller et al. (2017) that crowd valuations are noisy for smaller leagues, since in about 14% of the pairs the value does not change from one season to the next.

8.4 Implications for practice

For a club or an agent, the findings suggest caution. The archetypes provide a compact and interpretable description of the profile of a player and of how it changes, which may be useful for scouting and for monitoring the development of players, and the memberships of players in them show that profiles are fuzzy and change. As a signal for valuation or for identifying undervalued players, however, membership transitions cannot be recommended on this evidence: where an association is found, it is small, it depends on the tuning of the clustering and on the random draw, and it does not reliably improve prediction over a baseline of age, playing time, recent form and goals and assists, which is a more important determinant of changes in value than any of the transition variables. A practitioner who wants to use the approach should treat the memberships as noisy estimates, repeat the clustering with different seeds, report results for several settings, and not interpret a single membership change as a firm signal.

8.5 Limitations

Measurement of the profile. FBref does not publish expected goals, expected assists or passing and ball-progression statistics for the Primeira Liga over this period, so the archetypes describe attacking output, defensive actions and positions but not the way in which players move the ball. Richer data would give richer archetypes and could change the transitions. Positions are coarse, with three labels and five common combinations, and the weights of the attribute groups were fixed instead of estimated, which is a departure from the model of D’Urso et al. (2022) that was made for the reasons given in Chapter 5. The noise distance follows Davé (1991) and not the printed formula of D’Urso et al. (2022), and the shrinkage of rates is an additional step. Each departure was necessary for a well-behaved solution on these data, but each may matter for the results.

Estimated regressors and random draws. Memberships are noisy estimates; start stability is about 0.5, the bootstrap standard errors are on average 19% larger than the conventional ones, and the bootstrap estimates are shifted towards zero relative to the original coefficients. The bootstrap corrects the standard errors but not the attenuation, so effects may be understated, and there is no measure of by how much. Because the clustering is stochastic, a single run is one draw; the 30-draw analysis is done for the main configuration at five fuzziness values, not for every specification, and the bootstrap, which takes the original run as given, does not include the dependence of the original estimate on its draw.

Specification search. The conclusion depends on tuning parameters, and the thesis explored many specifications after the main configuration had been fixed. The main configuration is reported as the pre-specified one, and all specifications are reported, but the results at other settings, in particular $m = 1.4$ and 1.5 , have the status of exploratory findings. The proposal also envisaged selecting m with the Xie–Beni index, which was not done; the value that it would have chosen within the main configuration, $m = 1.4$, gives somewhat stronger results. The pipeline was also revised during the work: the alignment of the archetypes was corrected after a check showed inconsistent labels, and a data error in the merging of the FBref tables was repaired, each time because of a diagnostic and not an outcome, but the results of the earlier versions differed, which illustrates the sensitivity discussed above. No correction for the number of specifications leaves any result significant.

Data and sample. The valuation stage uses 1,361 pairs of 697 players, and the power to detect small effects is limited. Only players who stay in the league with sufficient playing time are observed in two consecutive seasons, so the findings do not apply to players who leave, who may be the ones whose value changes most. Market values are crowd estimates that move in steps, thinner for smaller leagues, and about 14% of the pairs show no change. The baseline omits contract length and international appearances, which are strong determinants of value in the literature (Franceschi et al., 2024; Poli et al., 2024), and it does not model the transfer fee. The link between FBref and Transfermarkt is made by name and birth year, with 96% of the player-seasons linked, a few wrong links removed by hand, and two players of the sample with

the same name merged by FBref.

Design. Independent clustering of the seasons ignores temporal dependence in the profiles; the number of ambiguous alignments is small at four archetypes (2 of 28 matches), and the alignment by profiles is consistent in all 120 runs, but the sensitivity of the results to the alignment method has not been examined beyond the comparison with the medoid-based version. The design is observational, and all findings are associations.

8.6 Directions for future research

Four directions follow. First, richer data, either from a provider with ball-progression and passing statistics or from event data, would allow archetypes that include how players move the ball. Second, the larger leagues provide many more transitions, and thus more power to detect small associations; a replication there, with the tuning decided in advance and the clustering repeated with many seeds, would show whether the dependence on fuzziness and on the draw is a feature of the Primeira Liga. Third, the noise in the memberships could be addressed directly, for example by methods that correct for errors in the regressors, by averaging the memberships over many random draws, or by models that estimate the profile and its change jointly instead of in two stages. Fourth, the association could be examined on longer horizons than one season, with contract length and international status among the controls, and for transfer fees in addition to market values.

Chapter 9

Conclusion

This thesis asked whether longitudinal transitions in the fuzzy cluster membership of football players have incremental explanatory power for changes in their market values, using the robust fuzzy C -medoids model for mixed data of D’Urso et al. (2022) on eight seasons of the Portuguese Primeira Liga. The analysis combined FBref statistics for 3,192 player-seasons of 1,499 outfield players with Transfermarkt valuations for 94% of them, clustered every season separately, aligned the archetypes across seasons, and related changes in memberships to changes in log market value for 1,361 pairs of consecutive seasons, with a bootstrap that accounts for the estimation of the memberships. The objective was explanatory and associational, and no causal claim is made.

The clustering identifies four stable and interpretable archetypes, defenders, central midfielders, wide attackers and strikers, and a small noise cluster. Players are fairly persistent in their archetype (75% remain), with defenders almost fixed and attacking players more mobile. The estimation of the attribute weights, a distinctive feature of the model, failed on these data and had to be replaced by weights fixed in proportion to the number of variables in each group, and the alignment of the archetypes across seasons had to use their profiles instead of their medoids.

In the main configuration, the membership transitions are associated with changes in market value beyond age, playing time, position, team form and goals and assists, and beyond all twelve performance statistics that enter the clustering: the bootstrap p -values are 0.015 against the full baseline and 0.079 against the core baseline, and the gains are small (0.3 to 0.6 percentage points of adjusted R^2 and 0.1 to 0.3 points of out-of-sample R^2). The association is concentrated in the shift from the defender profile to the wide-attacker profile and is not reproduced by a linear summary of the same statistics. It is, however, weak and specification-dependent: it is absent for near-crisp memberships and for five archetypes, it varies by orders of magnitude between random draws of the clustering (it is found against the full baseline in about two thirds of the draws and against the core baseline in about a third), a typical draw brings no improvement in out-of-sample prediction over the core baseline, and no result survives a correction for the number of specifications examined. The conclusion is therefore that membership transitions carry, at most, a small amount of information about changes in market value that is not captured by the performance statistics, and that robust incremental explanatory power cannot be claimed.

The thesis contributes a longitudinal application of a mixed-data fuzzy clustering model to a league outside the big five, a documented solution to the failure of weight estimation and to the inconsistency of medoid-based alignment, a bootstrap that accounts for estimated memberships,

and a full report of how the conclusion varies across 120 specifications and 30 random draws. Its limitations are the restricted set of statistics available, the noise in the memberships and in market values, the limited sample, and a specification search after the main configuration was fixed. For practitioners, the archetypes are a useful description but the transitions are not yet a basis for valuation. For research, replication in larger leagues with the tuning fixed in advance and the clustering repeated over many seeds, and methods that treat the noise in memberships directly, are the natural next steps.

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Appendix A

Programming code

All analyses were carried out in Python. The programs are submitted as separate files in a ZIP archive. They are listed below in the order in which they are run; each reads the outputs of the previous ones from a common working folder. All random numbers are generated from fixed seeds. The FBref and Transfermarkt data are collected by the programs and are not redistributed.

The estimated-weights results of Appendix B are reproduced by `estimated_weights_check.py`; the options that it uses are part of the `model` module (`fixed_weights=None`, the scaling `norm` set to `max` or `mean`, and the noise rule `dave`). The programs require Python 3 with `numpy`, `pandas`, `scipy`, `scikit-learn`, `statsmodels`, `linearmodels`, `joblib`, `matplotlib`, `soccerdata`, `requests`, `beautifulsoup4`, `rapidfuzz` and `unidecode`.

Table A.1: Programming files

File	Purpose
<code>test_fbref_collection.v2.py</code>	Registers the Primeira Liga in <code>soccerdata</code> , collects four FBref tables (standard, shooting, playing time, miscellaneous) for 2017/18 to 2024/25 and merges them.
<code>collect_tm.v2.py</code>	Collects Transfermarkt squad lists for each club and season, links FBref players to Transfermarkt players within each season, downloads the market value histories and extracts the end-of-season values.
<code>build_panel.py</code>	Removes rows that cannot be matched (same-name players at one club), merges the rows of players with several clubs in a season, applies the sample filters, builds the per-90 statistics, shrunk rates and position indicators, and scales the variables.
<code>make_analysis_panel.py</code>	Attaches market values, resolves players linked to two spellings, defines one identity per player and constructs the outcome for consecutive-season pairs.
<code>crosswalk_audit.py</code>	Lists the weakest FBref–Transfermarkt links for manual review.
<code>fcmd_mdnc.py</code>	Module with the clustering model: type-specific dissimilarities, the fuzzy <i>C</i> -medoids algorithm with noise cluster, fixed or estimated weights, the Xie–Beni index and the selection routines.
<code>cluster_one_season.py</code>	Development check of the clustering model on a single season.
<code>align_and_select.py</code>	Clusters every season, aligns the archetypes across seasons with the Hungarian algorithm on the distance between their profiles, builds the membership transitions and the diagnostics used to choose <i>K</i> .
<code>pairs_lib.py</code>	Module that builds the table of season pairs and the baselines, and the tests used by the regression programs.
<code>regression.py</code>	Main regressions: baselines, tests, in-sample and cross-validated explanatory power.
<code>bootstrap.py</code>	Two-stage bootstrap of both stages, resumable, for any configuration.
<code>robustness.py</code>	Sample and outcome variations, fixed and random effects and the Hausman test.
<code>sensitivity_and_benchmark.py</code>	Sensitivity to ρ and m and the principal component benchmark.
<code>spec_grid.py</code>	Re-runs clustering, alignment and tests for the 120 specifications of the grid.
<code>alignment_check.py</code>	Checks the consistency of the archetype labels across seasons in all 120 clustering runs.
<code>estimated_weights_check.py</code>	Fits the model with estimated attribute weights for one season in four variants (Appendix B).
<code>seed_sensitivity.py</code>	Repeats the clustering stage with 30 random seeds per fuzziness value and the key test each time.
<code>extra_results.py</code>	Complete coefficient tables and the baseline without performance statistics.
<code>make_tables_figures.py</code>	Produces the tables and figures of the thesis.

Appendix B

Estimated attribute weights

Table B.1 reports the attribute weights that the estimation of D’Urso et al. (2022) produced for the 2023/24 season ($n = 406$ players, $m = 1.3$, $\rho = 0.5$, ten random starts) in the four variants that were tried: distances scaled by their maximum, as in the original paper, or by their mean, with raw or shrunk success rates. This season was used to develop the model, and the other seasons were not examined in this way. The values were recomputed on the final panel and agree with those of the development runs to the third decimal.

Table B.1: Estimated attribute weights, season 2023/24

Scaling	Rates	K	Performance	Success rates	Position	Noise	XB	Degenerate
Max	Raw	3	0.038	0.959	0.003	5.2%	0.21	yes
Max	Raw	4	0.031	0.967	0.002	4.7%	0.22	yes
Max	Raw	5	0.000	0.000	1.000	0.0%	0.00	yes
Max	Shrunk	3	0.312	0.601	0.088	34.0%	0.34	no
Max	Shrunk	4	0.238	0.704	0.059	21.4%	0.86	no
Max	Shrunk	5	0.000	0.000	1.000	0.0%	0.00	yes
Mean	Raw	3	0.151	0.742	0.107	6.2%	0.42	no
Mean	Raw	4	0.091	0.038	0.871	16.0%	0.08	yes
Mean	Raw	5	0.000	0.000	1.000	0.0%	0.00	yes
Mean	Shrunk	3	0.128	0.109	0.762	4.4%	0.14	no
Mean	Shrunk	4	0.052	0.044	0.905	0.0%	0.05	yes
Mean	Shrunk	5	0.053	0.046	0.901	0.0%	6.53	yes

Note. Weights sum to one. Noise is the share of players whose largest membership is in the noise cluster and XB the Xie–Beni index. A solution is called degenerate if one attribute group has a weight above 0.8, if the Xie–Beni index is below 10^{-6} , or if more than half of the players are in the noise cluster. A weight of one on the positions with a noise share of zero is the case in which the clusters reproduce the position labels exactly. With the maximum, the two success rates dominate; with the mean, positions dominate. The weights used in the thesis are fixed at 0.501, 0.224 and 0.275 (Section 5.1). The table is reproduced by `estimated_weights_check.py`.

Appendix C

Coefficient tables

Tables C.1 and C.2 report all coefficients of the extended models for the main configuration, with cluster-robust standard errors (by player) that treat the memberships as observed. Statistics are in the scaled form of Section 4.3, in units of their average absolute deviation. The numbering of the archetypes is inherited from the first season and differs between runs. In the main configuration ($m = 1.3$) archetype 1 is the defenders (the reference), 2 the strikers, 3 the wide attackers and 4 the central midfielders; the same holds at $m = 1.4$ (Table C.3). At $m = 1.5$ the defenders are again archetype 1, and archetype 2 is the wide attackers, 3 the central midfielders and 4 the strikers.

Table C.1: Core baseline with membership changes, $m = 1.3$ (main configuration)

Variable	Coef.	SE	p
Age	-0.162	0.036	0.000
Age squared	0.002	0.001	0.005
Log minutes (level)	0.188	0.025	0.000
Change in log minutes	0.372	0.022	0.000
Defender	0.013	0.031	0.673
Forward	-0.042	0.042	0.312
Team points per match (level)	0.006	0.026	0.815
Change in team points per match	0.288	0.033	0.000
Change of club within the league	-0.012	0.036	0.728
Non-penalty goals per 90 (level)	0.069	0.014	0.000
Non-penalty goals per 90 (change)	0.061	0.013	0.000
Assists per 90 (level)	0.003	0.013	0.817
Assists per 90 (change)	0.023	0.012	0.064
Pair starting 2018/19	-0.131	0.046	0.004
Pair starting 2019/20	0.302	0.053	0.000
Pair starting 2020/21	-0.005	0.045	0.910
Pair starting 2021/22	-0.002	0.052	0.964
Pair starting 2022/23	0.145	0.050	0.004
Pair starting 2023/24	0.079	0.048	0.097
Change in membership of archetype 2	-0.036	0.102	0.723
Change in membership of archetype 3	-0.216	0.083	0.010
Change in membership of archetype 4	-0.104	0.072	0.147
Change in noise membership	-0.141	0.132	0.285

Table C.2: Full baseline with membership changes, $m = 1.3$ (main configuration)

Variable	Coef.	SE	p
Age	-0.159	0.036	0.000
Age squared	0.002	0.001	0.006
Log minutes (level)	0.193	0.027	0.000
Change in log minutes	0.375	0.024	0.000
Defender	0.049	0.039	0.210
Forward	-0.030	0.050	0.549
Team points per match (level)	0.006	0.027	0.818
Change in team points per match	0.296	0.033	0.000
Change of club within the league	-0.011	0.037	0.763
Non-penalty goals per 90 (level)	0.050	0.039	0.203
Non-penalty goals per 90 (change)	0.033	0.030	0.267
Assists per 90 (level)	0.002	0.015	0.883
Assists per 90 (change)	0.025	0.013	0.052
Pair starting 2018/19	-0.122	0.047	0.009
Pair starting 2019/20	0.312	0.056	0.000
Pair starting 2020/21	0.012	0.047	0.803
Pair starting 2021/22	0.008	0.056	0.890
Pair starting 2022/23	0.175	0.054	0.001
Pair starting 2023/24	0.113	0.053	0.033
Level of shots p90	-0.015	0.057	0.795
Change in shots p90	0.064	0.045	0.160
Level of sot p90	0.050	0.073	0.489
Change in sot p90	-0.024	0.050	0.624
Level of crosses p90	-0.003	0.011	0.766
Change in crosses p90	0.010	0.016	0.520
Level of interceptions p90	0.017	0.018	0.345
Change in interceptions p90	0.001	0.019	0.960
Level of tackles won p90	0.007	0.015	0.608
Change in tackles won p90	0.008	0.013	0.569
Level of fouls p90	0.008	0.015	0.611
Change in fouls p90	-0.010	0.015	0.528
Level of fouled p90	0.023	0.018	0.218
Change in fouled p90	0.049	0.022	0.024
Level of offsides p90	0.003	0.014	0.851
Change in offsides p90	-0.009	0.016	0.569
Level of sot rate shr	-0.046	0.031	0.134
Change in sot rate shr	0.005	0.024	0.830
Level of g sh shr	0.022	0.030	0.458
Change in g sh shr	0.027	0.024	0.276
Change in membership of archetype 2	-0.114	0.113	0.313
Change in membership of archetype 3	-0.308	0.089	0.001
Change in membership of archetype 4	-0.140	0.072	0.053
Change in noise membership	-0.239	0.161	0.138

Table C.3: Core baseline with membership changes, $m = 1.4$

Variable	Coef.	SE	p
Age	-0.160	0.036	0.000
Age squared	0.002	0.001	0.005
Log minutes (level)	0.186	0.025	0.000
Change in log minutes	0.377	0.022	0.000
Defender	0.013	0.031	0.676
Forward	-0.043	0.042	0.298
Team points per match (level)	0.006	0.026	0.815
Change in team points per match	0.286	0.033	0.000
Change of club within the league	-0.013	0.036	0.715
Non-penalty goals per 90 (level)	0.069	0.014	0.000
Non-penalty goals per 90 (change)	0.057	0.013	0.000
Assists per 90 (level)	0.002	0.013	0.871
Assists per 90 (change)	0.017	0.012	0.151
Pair starting 2018/19	-0.137	0.045	0.003
Pair starting 2019/20	0.294	0.054	0.000
Pair starting 2020/21	0.006	0.046	0.896
Pair starting 2021/22	-0.004	0.051	0.938
Pair starting 2022/23	0.140	0.051	0.006
Pair starting 2023/24	0.076	0.048	0.113
Change in membership of archetype 2	-0.061	0.105	0.563
Change in membership of archetype 3	-0.258	0.088	0.003
Change in membership of archetype 4	-0.123	0.076	0.105
Change in noise membership	-0.035	0.147	0.810